Differential Geometry

December 2022

Contents

Part I

Differential Geometry

Chapter 1

Foundational Differential Geometry

Introduction

Differential Geometry is a mathematical discipline that studies the geometry of smooth shapes and smooth spaces, otherwise known as smooth manifolds. Apart from mathematicians, knowledge of differential geometry is also common among physicists, thanks to the success of Einstein's highly geometric theory of gravitation and also as we see in the discovery of differential geometric underpinnings of modern gauge theory and string theory.

1.1 Differentiable Manifolds

A differentiable manifold is a topological space with defined co-ordinates that allow basic notions of differentiability. Manifolds of arbitrarily high dimension play an important role in many physical theories. For example, in Einstein's General Theory of Relativity, spacetime is taken to be a 4-dimensional manifold, whereas, in classical mechanics, the set of all possible locations and orientations of a rigid body is manifold of dimension 6, and the phase space[1]of a system of N Newtonian particles moving in a 3-dimensional space is a manifold of dimension 6N.

In this section we give basic definitions and properties of smooth manifolds and of smooth maps between manifolds. Initially the only way to verify that a space is a manifold is to exhibit a collection of C^∞ compatible charts covering the space. Later, we describe a set of sufficient conditions under which a quotient topological space becomes a manifold, giving us a second way to construct manifolds.

Definition 1.1.1 (Topological manifold)**.** An n-dimensional topological manifold is a paracompact Hausdorff space, say M, such that every point $p \in M$ is contained in some open set U_p that is homeomorphic to an open subset of the Euclidean space $\mathbb{R}^n$. The integer n is referred to as the **dimension** of M, and we denote it by $\dim(M)$.

Example 1.1.1. *The n-sphere*

$$S^n := \{(x^1, \ldots, x^n) \in \mathbb{R}^{n+1} : \Sigma(x^i)^2 = 1\}$$

is a topological manifold of dimension n .

If M_1 and M_2 are topological manifolds of dimension n_1 ans n_2 respectively, then $M_1 \times M_2$, with product topology, is a topological manifold of dimension $n_1 + n_2$. Such a manifold is called a product manifold. Indeed, the required homeomorphisms are constructed in the obvious way from those defined on M_1 and M_2 :

[1]In dynamical system theory, a phase space is a space in which all possible states of a system are represented, with each possible state corresponding to one unique point in the phase space.

If $\phi_p : U_p \subset M_1 \to V_{\phi(p)} \subset \mathbb{R}^{n_1}$ and $\psi_q : U_q \subset M_2 \to V_{\psi(q)} \subset \mathbb{R}^{n_2}$ are homeomorphisms, then $U_p \times U_q$ is a neighbourhood of (p, q) and we have a homeomorphism

$$\phi_p \times \psi_q : U_p \times U_q \to V_{\phi(p)} \times V_{\psi(q)} \subset \mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \ ,$$

where $(\phi_p \times \psi_q)(x, y) := (\phi_p(x), \psi_q(y))$. That $M_1 \times M_2$ is Hausdorff and paracompact is easy to see.

Example 1.1.2. *The n-torus*

$$T^n := S^1 \times S^1 \times \cdots \times S^1 \ (n \text{ factors})$$

is a topological manifold of dimension n.

Definition 1.1.2. A topological space is called **locally path connected** if the topology has a basis consisting of path connected sets.

Remark 1.1.1. Every Manifold is locally path connected, hence, locally connected.

Lemma 1.1.1. *The path components of a manifold M are exactly the connected components of M. Thus, a manifold is connected if and only if it is path connected.*

Proof. In general, a manifold is not connected, but every manifold M is a disjoint union of connected manifolds which are just the connected components of M. Since, every manifold is locally connected, the connected components are also open. Also, the manifold M being locally path connected, the path connected components are also open. Since, every path connected component is contained in some connected component , it follows that M is path connected if and only if M is connected. Hence, the connected and the path connected components of M are same. □

1.1.1 Charts, Atlases and Smooth Structures

In this section we introduce the notion of charts and co-ordinate systems. The existence of such charts is what allows a well-defined notion of differentiable functions on a manifold or from one manifold to the other.

Definition 1.1.3. Let M be a set. A chart on M is a bijection of a subset $U \subset M$ onto an open subset of some Euclidean space $\mathbb{R}^n$.

A chart $\mathbf{x} : U \to \mathbf{x}(U) \subset \mathbb{R}^n$ is traditionally indicated by the pair $(U, \mathbf{x})$. The i-th coordinate functions is defined by $\mathbf{x}^\mathbf{i} = \mathbf{pr_i} \circ \mathbf{x}$, where $\mathbf{pr_i} : \mathbb{R}^n \to \mathbb{R}$ is projection onto the i-th factor given by $\mathbf{pr_i}(a^1, \ldots, a^n) = a^i$. If $p \in U$ and $\mathbf{x}(p) = 0 \in \mathbb{R}^n$, then we say that the chart is centered at p.

Definition 1.1.4. Let $\mathcal{A} = \{(U_\alpha, \mathbf{x}_\alpha)\}_{\alpha \in \mathcal{I}}$ be a collection of $\mathbb{R}^n$-valued charts on a set M. We call $\mathcal{A}$ an $\mathbb{R}^n$-valued **atlas of class** C^r if the following conditions are satisfied:

(i) $\bigcup_{\alpha \in \mathcal{I}} U_\alpha = M$

(ii) The sets of the form $\mathbf{x}_\alpha(U_\alpha \cap U_\beta)$ for all $\alpha, \beta \in \mathcal{I}$ are all open in $\mathbb{R}^n$.

(iii) Whenever $U_\alpha \cap U_\beta$ is non-empty, the map $\mathbf{x}_\beta \circ \mathbf{x}_\alpha^{-1} : \mathbf{x}_\alpha(U_\alpha \cap U_\beta) \to \mathbf{x}_\beta(U_\alpha \cap U_\beta)$ is a C^r diffeomorphism.

The maps $\mathbf{x}_\beta \circ \mathbf{x}_\alpha^{-1}$ in the definition are called **overlap maps** or **change of coordinate maps** or equivalently, **transition functions**.

Definition 1.1.5. Two C^r atlases $\mathcal{A}_1$ and $\mathcal{A}_2$ on M are said to be **equivalent** provided that $\mathcal{A}_1 \bigcup \mathcal{A}_2$ is also a C^r atlas for M. A C^r **differentiable structure** on M is an equivalence class of C^r atlases. A C^∞ differentiable structure on M will also be called a **smooth structure**.

Heneceforth, a smooth structure on a manifold M is a collection of smoothly equivalent smooth atlases[2].

The union of all the C^r atlases of an equivalence class is itself an atlas that is in the equivalence class and this atlas is said to be **maximal** with respect to partial ordering of atlases by inclusion. We thereby obtain a **1-1 correspondence** between the set of equivalence classes of C^r differentiable structures and the set of maximal C^r atlases.

A pair of charts, say, $(U, \mathbf{x})$ and $(V, \mathbf{y})$ are said to be C^r**-related** if either $U \cap V = \phi$ or both $\mathbf{x}(U \cap V)$ and $\mathbf{y}(U \cap V)$ are open and $\mathbf{x} \circ \mathbf{y}^{-1}$ and $\mathbf{y} \circ \mathbf{x}^{-1}$ are C^r maps. Thus an atlas is just a family of mutually C^r-related charts whose domains form a cover of the manifold. A chart $(U, \mathbf{x})$ is said to be **compatible** with a C^r atlas $\mathcal{A}$ if $\mathcal{A} \cup \{(U, \mathbf{x})\}$ is also a C^r atlas. This just means that $(U, \mathbf{x})$ is C^r-related to every chart in $\mathcal{A}$.The maximal C^r atlas determined by $\mathcal{A}$ is composed of all charts compatible with $\mathcal{A}$. Also a chart from the maximal atlas which defines the smooth structure is called **admissible**[3].

A **smooth coordinate ball** is a smooth coordinate domain whose image under a coordinate function is a ball in Euclidean space.

Let M be a smooth manifold. A set $B \subseteq M$ is a **regular coordinate ball** if there is a smooth coordinate ball $B' \supseteq \bar{B}$ and a smooth coordinate map(function) $\phi : B \to \mathbb{R}^n$ such that for some positive real nos $r < r'$,

$$\phi(B) = B_r(0), \;\; \phi(\bar{B}) = \bar{B}_r(0), \;\; \text{and } \phi(B') = B_{r'}(0).$$

Example 1.1.3. *The space $\mathbb{R}^n$ itself has an atlas consisting of the single chart $(\mathbf{id}, \mathbb{R}^n)$ where $\mathbf{id} : \mathbb{R}^n \to \mathbb{R}^n$ is just the identity map. This atlas determines a differentiable structure.*

Lemma 1.1.2. *Let M be a set with C^r structure given by an atlas $\mathcal{A} = \{(U_\alpha, \mathbf{x}_\alpha)\}_{\alpha \in \mathcal{I}}$. If $(U, \mathbf{x})$ and $(V, \mathbf{y})$ are charts compatible with $\mathcal{A}$ such that $U \cap V \neq \phi$, then the charts $(U \cap V, \mathbf{x}|_{U \cap V})$ and $(U \cap V, \mathbf{y}|_{U \cap V})$ are also compatible with $\mathcal{A}$ and hence are in the maximal atlas generated by $\mathcal{A}$. Furthermore, if $\mathcal{O}$ is an open subset of $\mathbf{x}(U)$ for some compatible chart $(U, \mathbf{x})$, then taking $V = \mathbf{x}^{-1}(\mathcal{O})$ we have that $(V, \mathbf{x}|_V)$ is also a compatible chart.*

Proof. Since the charts $(U, \mathbf{x})$ and $(V, \mathbf{y})$ are compatible with $\mathcal{A}$, it follows that $\mathbf{x} \circ \mathbf{x}_\alpha^{-1}$, $\mathbf{x}_\alpha \circ \mathbf{x}^{-1}$, $\mathbf{y} \circ \mathbf{x}_\alpha^{-1}$, $\mathbf{x}_\alpha \circ \mathbf{y}^{-1}$ are all C^r diffeomorphisms, then certainly the restrictions $\mathbf{x}|_{U \cap V} \circ \mathbf{x}_\alpha^{-1}$, $\mathbf{x}_\alpha \circ \mathbf{x}|_{U \cap V}^{-1}$, $\mathbf{y}|_{U \cap V} \circ \mathbf{x}_\alpha^{-1}$, $\mathbf{x}_\alpha \circ \mathbf{y}|_{U \cap V}^{-1}$ are also. It is easy to check that the natural domains of these maps are also open in $\mathbb{R}^n$. For example, the domain of $\mathbf{x}|_{U \cap V} \circ \mathbf{x}_\alpha^{-1}$ is $\mathbf{x}_\alpha(U_\alpha \cap U \cap V) = \mathbf{x}_\alpha(U_\alpha \cap U) \cap \mathbf{x}_\alpha(U_\alpha \cap V)$, where both $\mathbf{x}_\alpha(U_\alpha \cap U)$ and $\mathbf{x}_\alpha(U_\alpha \cap V)$ are open by definition of compatibility of $(U, \mathbf{x})$ and $(V, \mathbf{y})$ with $\mathcal{A}$. The other part follows similarly.

□

The family of sets which are the domains of charts from the maximal atlas form a basis for a topology on M, which we call the **topology induced by the** C^r **structure** on M, or simply the **manifold topology**, if the C^r structure is understood. Thus the open sets are precisely the empty set plus the arbitrary union of chart domains from the maximal atlas. Since, a C^r structure is determined by a C^r atlas, this topology is also sometimes called the **topology induced by the atlas**.

Remark 1.1.2. If $\mathcal{A}_1$ is a subatlas of $\mathcal{A}_2$, then they both induce the same topology.

Proposition 1.1.1. *Let M be a set with a C^r structure given by an atlas $\mathcal{A}$. Then, we have the following:*

[2]A smooth atlas on a topological manifold M is an atlas where each transition function is smooth.

[3]In this note, charts will be assumed admissible unless otherwise specified.

(i) If for every two distinct points $p, q \in M$, we have either that p, q are in disjoint domains U_α and U_β respectively, from the atlas, or they are both in a common chart domain, then the topology induced by the atlas is Hausdorff.

(ii) If $\mathcal{A}$ is countable or has a countable subatlas, then the topology induced by the atlas is second countable.

(iii) If the collection of chart domains $\{U_\alpha\}_{\alpha\in\mathcal{I}}$ from the atlas $\mathcal{A}$ is such that for every fixed $\alpha_0 \in \mathcal{I}$ the set $\{\alpha \in \mathcal{A} : U_\alpha \cap U_{\alpha_0} \neq \phi\}$ is at most countable, then the topology induced by the atlas is paracompact. Thus, if the condition holds and M is connected, then the topology induced by the atlas is second countable.

Proof. We leave the proof for later. $\square$

Definition 1.1.6. A **differentiable manifold** of class C^r is a set M together with a specified C^r structure on M such that the topology induced by the C^r structure is Hausdorff and paracompact.

In other words, an n-dimensional differentiable manifold of class C^r is a tuple $(M, \mathcal{A})$, where $\mathcal{A}$ is a maximal C^r atlas such that the topology induced by the atlas makes M a topological manifold. A differentiable manifold of class C^∞ is called a **smooth manifold**.

Now, perhaps it is already known that M is a topological manifold. Then, if further M is given a C^r structure, it is important to know whether this topology is same as the topology induced by the C^r structure. For this consideration, we have the following lemma:

Lemma 1.1.3. *Let (M, τ) be a topological space which also has a C^r atlas. If each chart of the atlas has open domain and is a homeomorphism with respect to this topology, then τ will be the same as the topology induced by the C^r structure.*

By the above lemma, one often defines differentiable manifolds as topological manifolds that have a C^r atlas consisting of charts that are homeomorphisms.

From now on, all manifolds will be considered smooth manifolds. And, a n-dimensional smooth manifold will be referred as an **"n-manifold"**.

It is certainly possible to have two different differentiable structures on the same topological manifold. For example, if $\phi \to \mathbb{R}^1 \to \mathbb{R}^1$ is given by $\phi(x) = x^3$, then $\{(\mathbb{R}^1, \phi)\}$ is a smooth atlas for $\mathbb{R}^1$, but the resulting smooth structure is different from the usual structure provided by the atlas $\{(\mathbb{R}^1, \mathbf{id})\}$. It is because the inverse of $x \mapsto x^3$ is not differentiable (in the usual sense) at the origin.

Now let us demonstrate how we can use stereographic projection to give charts on S^n: We can provide the $n-sphere$ $S^n \subset \mathbb{R}^{n+1}$ with a smooth structure using two charts (U_S, ψ_S) and (U_N, ψ_N). Here,

$$U_S = \{x = (x_1, \ldots, x_{n+1}) \in S^n : x_{n+1} \neq 1\},$$
$$U_N = \{x = (x_1, \ldots, x_{n+1}) \in S^n : x_{n+1} \neq -1\}$$

and $\psi_S : U_S \to \mathbb{R}^n$ (resp.$\psi_N : U_N \to \mathbb{R}^n$) is stereographic projection from the north pole $p_N = (0, \ldots, 1)$ (resp. south pole $p_S = (0, \ldots, -1)$). Explicitly, we have

$$\psi_S(x) = \tfrac{1}{(1-x_{n+1})}(x_1, \ldots, x_n) \in \mathbb{R}^n,$$
$$\psi_N(x) = \tfrac{1}{(1+x_{n+1})}(x_1, \ldots, x_n) \in \mathbb{R}^n$$

1.2 Smooth maps on a Manifold

Now that we have defined smooth manifolds, we are ready to define smooth maps between them. Using coordinate charts, one can transfer the notion of smooth maps from Euclidean spaces to manifolds. By the C^∞ compatibility of charts in an atlas, the smoothness of a map turns out to be independent of the choice of charts and is therefore well defined.

1.2.1 Smooth Functions on a Manifold

Definition 1.2.1. Let M be a smooth manifold of dimension n. **A function $f : M \to \mathbb{R}$ is said to be smooth at a point** p in M if there is a chart (U, ϕ) about p in M such that $f \circ \phi^{-1}$, a function defined on the open set $\phi(U)$ of $\mathbb{R}^n$ is C^∞ at $\phi(p)$. The function f is said to be smooth on M if it is C^∞ at every point of M.

1.2.2 Smooth Maps between Manifolds

Definition 1.2.2. Let N and M be manifolds of dimension n and m respectively. A continuous map $F : N \to M$ is C^∞ at a point p in N if there are charts (V, ψ) about $F(p)$ in M and (U, ϕ) about p in N such that the composition $\psi \circ F \circ \phi^{-1}$, a map from the open subset $\phi(F^{-1}(V) \cap U)$ of $\mathbb{R}^n$ to $\mathbb{R}^m$ is C^∞ at $\phi(p)$. The continuous map $F : N \to M$ is said to be *smooth* if it is C^∞ at every point of N.

1.3 Diffeomorphisms

let M and N be two smooth manifolds. M is said to be diffeomorphic to N if there is a smooth bijective map $f : M \to N$ such that f^{-1} is also smooth.

Example 1.3.1. *The topological manifolds RP^n and $S^n/\sim$ are diffeomorphic.*

Example 1.3.2. *If M is any smooth manifold and (U, ϕ) is a smooth coordinate chart on M, then $\phi : U \to \phi(U) \subseteq$ is a diffeomorphism, with identity map as its coordinate representation.*

Example 1.3.3. *Consider the following maps $F : \mathbb{B}^n \to \mathbb{R}^n$ and $G : \mathbb{R}^n \to \mathbb{B}^n$ defined by*

$$F(x) = \frac{x}{\sqrt{1-|x|^2}}, \qquad G(y) = \frac{y}{\sqrt{1+|y|^2}}.$$

These maps are clearly smooth and are inverses of each other. hence, they both are diffeomorphisms, and thus, $\mathbb{B}^n$ is diffeomorphic to $\mathbb{R}^n$.

Proposition 1.3.1. ***(Properties of diffeomorphisms):***

1. *Every composition of diffeomorphisms is a diffeomorphism.*
2. *Every finite product of diffeomorphisms between smooth manifolds is a smooth manifold.*
3. *Every diffeomorphism is a homeomorphism and an open map.*
4. *The restriction of a diffeomorphism to an open submanifold with or without boundary is a diffeomorphism onto its image.*
5. *"Diffeomorphic" is an equivalence relation on the class of all smooth manifolds with or without boundary.*

The following theorem is a weak version of *invariance of dimension*, which suffices for many purposes.

Theorem 1.3.1. ***(Diffeomorphism invariance of dimension).*** *A non empty smooth manifold of dimension m cannot be diffeomorphic to an n-dimensional smooth manifold unless m=n.*

Proof. Let M and N be two manifolds of dimension m and n respectively and let $F : M \to N$ be a diffeomorphism. Let $p \in M$ be arbitrary, and let (U, ϕ) and (V, ψ) be smooth coordinate charts containing p and $F(p)$ respectively. Then, (the restriction of) $\hat{F} = \psi \circ F \circ \phi^{-1}$ is a diffeomorphism from an open subset of $\mathbb{R}^m$ to an open subset of $\mathbb{R}^n$, so it follows that $m = n$. $\square$

A similar theorem holds in the case of invariance of boundaries.

Theorem 1.3.2. ***(Diffeomorphism invariance of the boundary).*** *Suppose M and N are smooth manifolds with boundary and $F : M \to N$ is a diffeomorphism. Then $F(\partial M) = \partial(N)$, and F restricts to a diffeomorphism from Int M to Int N.*

Now, an interesting question arises whether a given topological manifold with two different smooth structures are diffeomorphic. It turns out from works of James Munkres and Edwin Moise that every topological manifold with dimension less than or equal to 3 has a smooth structure that is unique upto diffeomorphism. But, this does not hold true for manifolds of dimension greater than 3. For manifolds of dimension ≥ 5, there are atmost finitely many smooth structures, in fact, in all dimensions greater than 3 there are compact topological manifolds that have no smooth structure at all. The wild thing happens with manifolds of dimension equal to 4. There are many simply connected closed 4-manifolds that admit infinitely many distinct smooth structures. There are no smooth 4-manifolds known to have finitely many smooth structures.

For open manifolds, things get even worse: For $n \neq 4$, the topological manifold $\mathbb{R}^n$ admits unique smooth structure. However, topological 4-manifold $\mathbb{R}^4$ admits uncountably many distinct smooth structures.

Let us give an example when $\mathbb{R}$ is endowed with two smooth structures. Let $\tilde{\mathbb{R}}$ denote the topological manifold $\mathbb{R}$, but endowed with a smooth structure defined by the global chart $\psi(x) = x^3$. It turns out that $\tilde{\mathbb{R}}$ is diffeomorphic to $\mathbb{R}$ with its standard smooth structure. We define a map $F : \mathbb{R} \to \tilde{\mathbb{R}}$ by $F(x) = x^{1/3}$. the coordinate representation of this map is $\hat{F}(t) = \psi \circ Id_{\mathbb{R}}^{-1}(t) = t$, which is clearly smooth. also, it is easy to see that the inverse of this coordinate representation is again the identity map, which is indeed smooth. Thus F is a diffeomorphism.

The problem of identifying all the smooth structures (if any) on topological-4-manifold is a current subject of active research.

Before we start our next topic, that is Partition of Unity, we provide the following lemma which will help appreciate the existence of such a notion:

Lemma 1.3.3. *Let M be a smooth manifold such that $M = U \cup V$, where U and V are open in M. Suppose $f_1 : U \to \mathbb{R}$ and $f_2 : V \to \mathbb{R}$ are smooth on M and $f_1 = f_2$ on $U \cap V$. Then $f : M \to \mathbb{R}$ is smooth where $f \equiv f_1$ on U and $f \equiv f_2$ on V.*

1.4 Partition of Unity

The above lemma corresponds to the gluing lemma for smooth maps defined on open subsets of a smooth manifold, but we cannot always expect to glue together smooth maps defined on closed subsets and obtain a smooth result. This is where partition of unity comes into play as it can be considered as a tool for "blending together" local smooth objects into global one without necessarily assuming that they agree on overlaps.

The following two theorems help in proving the existence of partition of unity:

Theorem 1.4.1. *Every topological manifold admits a locally finite open cover of pre-compact open sets.*

Proof. Will write soon.

□

Let us first give two definition before we propose our next theorem.

Definition 1.4.1. (Refinement of an open cover). Let $\mathcal{V}$ be an open cover of a topological space X. Then $\mathcal{U}$ is said to be the refinement of $\mathcal{V}$ if for all $U \in \mathcal{U}, \exists V \in \mathcal{V}$ such that $U \subset V$.

Definition 1.4.2. (Regular cover). Let M be a smooth manifold. A collection $\{B_i\}_{i\in I}$ is called a regular cover of M if -

i)$\{B_i\}_{i\in I}$is countable and locally finite.

ii) Each B_i is the domain of a smooth coordinate ball $\phi : B_i \to B_{r_3}(0)$.

iii) $\{U_i\}_{i\in I}$ still covers M, where $U_i = \phi^{-1}(B_{r_1}(0))$.

Theorem 1.4.2. *Any open cover of a smooth manifold admits a regular refinement.*

Proof. This also I will write soon(hopefully). □

From now, every construction in this section is based on the existence of smooth functions which is positive in a specified part of a manifold and identically zero in some other part. We begin by defining a smooth function on the real line that is zero for $t \le 0$ and positive for $t > 0$.

The function $f : \mathbb{R} \to \mathbb{R}$ is defined by

$$f(t) = \begin{cases} e^{-1/t} & \text{if } t > 0 \\ 0 & \text{if } t \le 0 \end{cases}$$

With some simple calculations, one can verify that the above function is smooth.

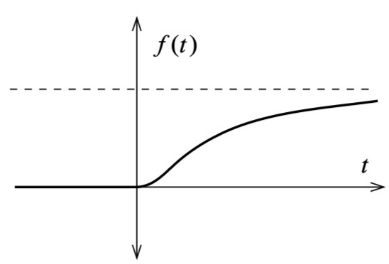

Figure 1.1: $f(t) = e^{-1/t}$

Lemma 1.4.3. *Given any real numbers r_1 and r_2 such that $r_1 < r_2$, there exists a smooth function $h : \mathbb{R} \to \mathbb{R}$ such that $h(t) \equiv 1$ for $t \le r_1$, $0 < h(t) < 1$ for $r_1 < t < r_2$, and $h(t) \equiv 0$ for $t \ge r_2$.*

Proof. Considering the above function f, we define another function $h : \mathbb{R} \to [0, 1]$ by

$$h(t) = \frac{f(r_2 - t)}{f(r_2 - t) + f(t - r_1)}$$

It is easy to see that the denominator is always positive for all t, since at least one of the expression $r_2 - t$ or $t - r_1$ is always positive. The desired properties of the function h follow those from those of f. □

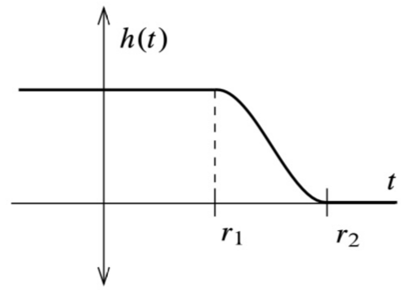

Figure 1.2: Cut off function $h(t)$

The function h with properties as mentioned in the previous lemma is usually called a **cut off function**.

Lemma 1.4.4. *Given any positive real numbers $r_1 < r_2$, there is a smooth function $H : \mathbb{R}^n \to \mathbb{R}$ such that $H \equiv 1$ on $\bar{B}_{r_1}(0)$, $0 < H(x) < 1$ for all $x \in B_{r_2}(0) \setminus \bar{B}_{r_1}(0)$, and $H \equiv 0$ on $\mathbb{R}^n \setminus B_{r_2}(0)$.*

Proof. Considering the function h of the previous lemma, we set $H(x) = h(|x|)$. Clearly, H is smooth on $\mathbb{R}^n \setminus \{0\}$, being a composition of two smooth functions. Also, H being identically equal to 1 on $B_{r_1}(0)$, it is smooth there too. $\square$

The above constructed function is an example of a **smooth bump function**, a smooth real valued function that is identically equal to 1 on a specified set and it vanishes outside a specified neighbourhood of that set.

Let f be any real valued or vector valued function on a topological space M, then **support of** f, denoted by *supp f*, is defined as the closure of set of all points where the function f does not vanish, that is

$$\text{supp } f = cl_M(f^{-1}(\mathbb{R}^*)), \text{ where } \mathbb{R}^* \text{ is the set of all non-zero reals.}$$

For example, in the above constructed function H, supp $H = \bar{B}_{r_2}(0)$. If the support of some function lies in an set U, then it is said that f is **supported in U**. A function supported on a compact set is said to be **compactly supported**, hence, every function defined on a compact set is compactly supported.

The following lemma shows one of the nicest consequences of existence of bump functions:

Lemma 1.4.5. *Let f be C^∞ function defined on a neighbourhood U of point p in a manifold M. then there is a C^∞ extension $\tilde{f}$ that agrees with f on a possibly smaller neighbourhood of p.*

Proof. To show this, we need to choose a bump function $\rho : M \to \mathbb{R}$ such that $\text{supp}(\rho) \subset U$ and there is an open neighbourhood V of p such that $V \subset U$ and ρ is identically equal to 1 on V.
We define the function $\tilde{f}$ as follows:

$$\tilde{f}(q) = \begin{cases} \rho(q)f(q) & \text{if q} \in U \\ 0 & \text{if q} \notin U \end{cases}$$

As a product of two C^∞ functions, $\tilde{f}$ is smooth in U. Also, since $\text{supp}(\rho)$ is compact, it is closed in $\mathbb{R}$. Hence, when $q \notin U$ there exists a open set containing q in which $\tilde{f}$ takes value 0 identically. Hence, $\tilde{f}$ is a smooth function on the entire manifold M. Finally, since $\rho \equiv 1$ on V, $\tilde{f}$ agrees with f on V. $\square$

Now, we define our main goal of this section, for which all the above constructions were done : Let M be a topological space and let $\mathcal{X} = (X_\alpha)_{\alpha \in A}$ be an arbitrary open cover cover of M indexed by a set A. A ***partition of unity subordinate to*** $\mathcal{X}$ is an indexed family $(\psi_\alpha)_{\alpha \in A}$ of continuous functions $\psi_\alpha : M \to \mathbb{R}$ with the following properties:

1. $0 \le \psi_\alpha(x) \le 1$ for all $\alpha \in A$ and all $x \in M$.
2. $\operatorname{supp} \psi_\alpha \subseteq X_\alpha$ for each $\alpha \in A$.
3. The family of supports $(\operatorname{supp} \psi_\alpha)_{\alpha \in A}$ is locally finite.
4. $\sum\limits_{\alpha \in A} \psi_\alpha(x) = 1$ for all $x \in M$.

Because of the local finiteness condition in (3), the sum in (4) has only finitely many non zero terms, so there is no issue with convergence.

Theorem 1.4.6. ***(Existence of smooth Partition of Unity).*** *Suppose M is a smooth manifold with or without boundary, and $\mathcal{X} = (X_\alpha)_{\alpha \in A}$ is any indexed open cover of M. Then there exists a smooth partition of unity subordinate to $\mathcal{X}$.*

Proof. For simplicity, we work with the case of manifold without boundary. We know, that every open subset of M is again a smooth manifold, hence, each set X_α is a smooth manifold. It is also easy to see every smooth manifold has a basis $\mathcal{B}_\alpha$ of regular coordinate balls (it is a direct consequence of theorem 1.4.1.) and thus, $\mathcal{B} = \bigcup\limits_\alpha \mathcal{B}_\alpha$ is a basis for the topology on M. It follows from theorem 1.4.2 that $\mathcal{X}$ has a countable, locally finite refinement $\{B_i\}$ containing the elements of $\mathcal{B}$. And, since M is a topological space, it follows that the cover $\{\bar{B}_i\}$ is also locally finite.

The fact that B_i is a regular coordinate ball in some X_α guarantees that there is a regular coordinate ball $B_i' \subseteq X_\alpha$ such that $B_i' \supseteq \bar{B}_i$, and a smooth coordinate map $\phi_i : B_i' \to \mathbb{R}^n$ such that $\phi_i(\bar{B}_i) = \bar{B}_{r_2}(0)$ and $\phi_i(B_i') = B_{r_3}(0)$ for some $r_2 < r_3$. For each i, define a function $f_i : M \to \mathbb{R}$ by

$$f_i = \begin{cases} H_i \circ \phi_i & \text{on } B_i', \\ 0 & \text{on } M \setminus \bar{B}_i, \end{cases}$$

where $H_i : \mathbb{R}^n \to \mathbb{R}$ is a smooth function that is positive in $B_{r_2}(0)$ and zero elsewhere as constructed in lemma 1.4.4. On the set $B_i' \setminus \bar{B}_i$ where the two definitions overlap, both yield the zero function, so f_i is well defined and smooth, and $\operatorname{supp} f_i = \bar{B}_i$.

Define $f : M \to \mathbb{R}$ by $f(x) = \sum\limits_i f_i(x)$. Because of the local finiteness of the cover $\{\bar{B}_i\}$, this sum has only finitely many non-zero terms in a neighbourhood of each point and thus defines a smooth function. Because each f_i is non-negative everywhere and positive on B_i, it follows that $f(x) > 0$ everywhere on M. Thus, the functions $g_i : M \to \mathbb{R}$ defined by $g_i(x) = f_i(x)/f(x)$ are also smooth. And it is clear from the definition that $0 \le g_i \le 1$ and $\sum\limits_i g_i \equiv 1$.

Finally, we need to reindex our functions so that they are indexed by the same set A as our open cover. Because our cover $\{B_i'\}$ is a refinement of $\mathcal{X}$, for each i, we can choose index $a(i) \in A$ such that $B_i' \subseteq X_{a(i)}$. For each $\alpha \in A$, define $\psi_\alpha : M \to \mathbb{R}$. For each $\alpha \in A$, define $\psi_\alpha : M \to \mathbb{R}$ by

$$\psi_\alpha = \sum_{i : a(i) = \alpha} g_i.$$

If there are no indices i such that $a(i) = \alpha$, then this sum should be interpreted as the zero function. and, we have

$$\text{supp } \psi_\alpha = \overline{\bigcup_{i:a(i)=\alpha} B_i} = \bigcup_{i:a(i)=\alpha} \bar{B}_i \subseteq X_\alpha.$$

Each ψ_α is a smooth function that $0 \leq \psi_\alpha \leq 1$. Moreover, the family of supports $(\text{supp } \psi_\alpha)_{\alpha \in A}$ is still locally finite, and $\sum_\alpha \psi_\alpha \equiv \sum_i g_i \equiv 1$, and so this is our desired partition of unity. □

1.5 The Tangent Space

By definition, the tangent space to a manifold at a point is the vector space of derivations at that point. Every smooth map of manifolds induces a linear map, called its *differential*, of tangent spaces at corresponding points. In particular, the *differential* of a map between manifolds is a generalization of the derivative of a map between Euclidean spaces.

We know that any problem is "good" if one can reduce it down to a linear algebra problem. Manifolds produce a great insight in this sense since a basic principle in manifold theory is the linearization principle, according to which a manifold can be approximated near a point by its tangent space at that point, and a smooth map can be approximated by the differential of the map. In this way, one reduces down a topological problem into a linear problem, which indeed eases out computations. A good example of the linearization principle is the *Inverse Function Theorem*, which reduces the local invertibility of a smooth map to the invertibility of its differential at a point.

Now, for any point p in an open set U of $\mathbb{R}^n$, there are two equivalent ways to define a tangent vector at p.

1. as an arrow, which can be represented by a column vector;
2. as a point derivative of C_p^∞, the algebra of germs of C^∞ functions at p.

Both definitions generalize to a manifold. The arrow approach deals with first choosing a chart (U, ϕ) at p and then decreeing a tangent vector at p to be an arrow at $\phi(p)$ in $\phi(U)$. Though being easier to visualize, this approach is quite difficult to work with since a different chart $(V.\psi)$ at p would give rise to different set of tangent vectors and one would then have to decide how to identify the arrows at $\phi(p)$ in $\phi(U)$ with the arrows at $\psi(p)$ in $\psi(V)$.

So, we adopt to the second approach as mentioned above.

1.5.1 The Tangent Space at a Point

A **germ** of a C^∞ function at p in M is an equivalence class of C^∞ functions defined in a neighbourhood of p in M, two such functions being equivalent if they agree on some, possibly smaller neighbourhood of p. The set of germs of C^∞ real-valued functions at p in M is denoted by $C_p^\infty(M)$. The addition and multiplication of functions make C_p^∞ into a ring; with scalar multiplication by real nos., the same becomes an algebra over $\mathbb{R}$.

A ***derivation*** at a point p in a manifold M, or a one point-derivation of C_p^∞, is a linear map $D : C_p^\infty(M) \to \mathbb{R}$ such that

$$D(fg) = (Df)g(p) + f(p)D(g).$$

Definition 1.5.1. A *tangent vector* at a point p in a manifold M is a derivation at p.

In the case of $\mathbb{R}^n$, the tangent space at a point p in $\mathbb{R}^n$ is in $1-1$ correspondence with the set of all directional derivatives at the point p. It is because corresponding to every tangent vector v at a point p in $\mathbb{R}^n$, the directional derivative at p along that vector v gives a map of real vector spaces $D_v : C_p^\infty \to \mathbb{R}$ Just as in the case of $\mathbb{R}^n$, the tangent vectors at p form a vector space $T_p(M)$, called the *tangent space* of M at p.

Remark 1.5.1. (Tangent space to an open subset). If U is an open subset containing the point p in M, then the algebra $C_p^\infty(U)$ of germs of C^∞ functions in U at p is same as $C_p^\infty(M)$. Hence, $T_pU = T_pM$.

1.5.2 The Differential of a Map

Let $F : N \to M$ be a C^∞ map between two manifolds. At each point $p \in N$, the map F induces a linear map of tangent spaces called its *differential* at p,

$$F_* : T_pN \to T_{F(p)}M$$

as follows. If $X_p \in T_pN$, then $F_*(X_p)$ is the tangent vector in $T_{F(p)}M$ defined by:

$$(F_*(X_p))f = X_p(f \circ F) \in \mathbb{R} \text{ for } f \in C^\infty_{F(p)}(M).$$

Here, f is a germ at $F(p)$, represented by a C^∞ function in a neighbourhood of $F(p)$.

As we can observe, the tangent vectors, which are the derivations are acting as an operator on C^∞ functions, from the algebra of germs at $F(p)$, or as we say $C^\infty_{F(p)}$. It is because for any manifold N, there is a $1-1$ correspondence between the tangent space of N at a point, say, $p \in N$, that is T_pN and the set of all directional derivatives at the point p, say $\mathcal{D}_p$ via the map $v \mapsto D_v$.

Example 1.5.1. *(Differential of a map between Euclidean Spaces).*
Suppose $F : \mathbb{R}^n \to \mathbb{R}^m$ is smooth and p is a point in $\mathbb{R}^n$. Let $x^1, x^2, \ldots, x^n$ be the coordinates on $\mathbb{R}^n$ and $y^1, y^2, \ldots, y^m$ be the coordinates on $\mathbb{R}^m$. Then the tangent vectors $\frac{\partial}{\partial x^1}|_p, \ldots, \frac{\partial}{\partial x^n}|_p$ form a basis for the tangent space $T_p(\mathbb{R}^n)$ [It is easily checked that $\frac{\partial}{\partial x^i}$ satisfies the derivation property and hence, is a tangent vector at p] and $\frac{\partial}{\partial y^1}|_{F(p)}, \ldots, \frac{\partial}{\partial y^n}|_{F(p)}$ form a basis for the tangent space $T_{F(p)}(\mathbb{R}^m)$. The linear map $F_* : T_p(\mathbb{R}^n) \to T_{F(p)}(\mathbb{R}^m)$ is described by a matrix relative to these two bases:

$$F_*\Big(\frac{\partial}{\partial x^j}\Big|_p\Big) = \sum_k a_j^k \Big(\frac{\partial}{\partial y^k}\Big|_{F(p)}\Big), \qquad a_j^k \in \mathbb{R}.$$

Evaluating both sides on y^i, we get the matrix of F relative to the above two bases as $[\frac{\partial F^i}{\partial x^j}(p)]$ which is precisely the Jacobian matrix of the derivative of F at p. Thus, the differential of a map between manifolds generalizes the derivative of a map between Euclidean spaces.

1.5.3 The Chain Rule

Let $F : N \to M$ and $G : M \to P$ be smooth maps of manifolds, and $p \in N$. The differentials of F at p and G at $F(p)$ are linear maps

$$T_pN \xrightarrow{F_{*,p}} T_{F(p)}M \xrightarrow{G_{*,F(p)}} T_{G(F(p))}P$$

Theorem 1.5.1. *If $F : N \to M$ and $G : M \to P$ are smooth maps of manifolds and $p \in N$, then*

$$(G \circ F)_{*,p} = G_{*,F(p)} \circ F_{*,p}$$

Proof. Let $X_p \in T_pN$ and let f be a smooth function at $G(F(p))$ in P. Then,

$$((G \circ F)_* X_p)f = X_p(f \circ G \circ F)$$

and

$$((G_* \circ F_*)X_p)f = (G_*(F_*X_p))f = (F_*X_p)(f \circ G) = X_p(f \circ G \circ F).$$

□

Remark 1.5.2. The differential of the identity map $\mathbb{1}_M : M \to M$ at any point $p \in M$ is the identity map

$$\mathbb{1}_{T_pM} : T_pM \to T_pM \text{ at any point } p \in M,$$

because

$$((\mathbb{1}_M)_*X_p)f = X_p(f \circ \mathbb{1}_M) = X_pf,$$

for any $X_p \in T_pM$ and $f \in C_p^\infty(M)$.

Corollary 1.5.1. *If $F : N \to M$ is a diffeomorphism of manifolds and $p \in N$, then $F_* : T_pN \to T_{F(p)}M$ is an isomorphism of vector spaces.*

Proof. To say that F is a diffeomorphism means that it has a differentiable inverse $G : M \to N$ such that $G \circ F = \mathbb{1}_{\mathbb{N}}$ and $F \circ G = \mathbb{1}_{\mathbb{M}}$. By the chain rule,

$$(G \circ F)_* = G_* \circ F_* = (\mathbb{1}_N)_* = \mathbb{1}_{T_pN}$$

$$(F \circ G)_* = F_* \circ G_* = (\mathbb{1}_M)_* = \mathbb{1}_{T_pM}$$

Hence, F_* and G_* are isomorphisms. □

Corollary 1.5.2. ***(Invariance of Dimension).*** *If an open set $U \subset \mathbb{R}^n$ is diffeomorphic to an open set $V \subset \mathbb{R}^m$, then $n = m$.*

Proof. Let $F : U \to V$ be a diffeomorphism and let $p \in U$. Then, by the above corollary, $F_{*,p} : T_pU \to T_{F(p)}V$ is an isomorphism of vector spaces. Since, they are vector space isomorphisms, $T_pU \simeq \mathbb{R}^n$ and $T_{F(p)}M \simeq \mathbb{R}^m$, and thus, we must have $n = m$. □

Now, given a coordinate neighbourhood $(U, \phi) = (U, x^1, \ldots, x^n)$ about a point p in a manifold M, and $r^1, \ldots, r^n$ being the standard coordinates on $\mathbb{R}^n$, we have:

$$x^i = r^i \circ \phi : U \to \mathbb{R}.$$

If f is a smooth function in a neighbourhood of p, we set

$$\frac{\partial}{\partial x^i}|_p f = \frac{\partial}{\partial r^i}|_{\phi(p)}(f \circ \phi^{-1}) \in \mathbb{R}$$

If (U, ϕ) be a chart about a point p in a manifold M, then, $\phi : U \to \mathbb{R}^n$ is a diffeomorphism onto its image, and thus, the differential,

$$\phi_* : T_pM \to T_{\phi(p)}\mathbb{R}^n$$

is a vector space isomorphism. In particular, the tangent space T_pM has the same dimension n as the manifold M.

Proposition 1.5.1. *Let $(U, \phi) = (U, x^1, \ldots, x^n)$ be a chart about a point p in a manifold M. Then,*

$$\phi_*(\frac{\partial}{\partial x^i}|_p) = \frac{\partial}{\partial r^i}|_{\phi(p)}.$$

Proof. For any $f \in C^\infty_{\phi(p)}(\mathbb{R}^n)$,

$$\begin{aligned}
\phi_*\Big(\frac{\partial}{\partial x^i}\Big|_p\Big)f &= \frac{\partial}{\partial x^i}\Big|_p(f\circ\phi) \quad \text{(definition of } \phi_*) \\
&= \frac{\partial}{\partial r^i}\Big|_{\phi(p)}(f\circ\phi\circ\phi^{-1}) \quad \text{(definition of } \frac{\partial}{\partial x^j}|_p) \\
&= \frac{\partial}{\partial r^i}\Big|_{\phi(p)}f
\end{aligned}$$

□

Proposition 1.5.2. *If $(U,\phi) = (U, x^1, \ldots, x^n)$ is a chart containing p, then the tangent space T_pM has basis*

$$\tfrac{\partial}{\partial x^1}|_p, \ldots, \tfrac{\partial}{\partial x^n}|_p$$

Proof. An isomorphism of vector spaces carries a basis to basis . By the above proposition, $\phi_* : T_pM \to T_{\phi(p)}(\mathbb{R}^n)$ maps $\frac{\partial}{\partial x^1}|_p, \ldots, \frac{\partial}{\partial x^n}|_p$ to $\frac{\partial}{\partial r^1}|_{\phi(p)}, \ldots, \frac{\partial}{\partial r^n}|_{\phi(p)}$, which is a basis for the tangent space $T_{\phi(p)}(\mathbb{R}^n)$. Therefore, $\frac{\partial}{\partial x^1}|_p, \ldots, \frac{\partial}{\partial x^n}|_p$ is a basis for T_pM. □

1.5.4 Immersions and Submersions

When we are given a smooth map between Euclidean spaces or two manifolds, our main goal obviously is to analyze the given map at best. That's where differentials come into play, because these linear maps best approximates our given C^∞ map at a point. Here, two cases, *i.e. of immersions and submersions* are especially important.

A C^∞ map $F : N \to M$ is said to be:

1. an *immersion* at $p \in N$ if its differential $F_{*,p} : T_pN \to T_pM$ is injective;

2. a *submersion* at $p \in N$ if its differential $F_{*,p} : T_pN \to T_pM$ is surjective.

We call F an *immersion* if F is an immersion at every $p \in N$ and a *submersion* if F is a submersion at every $p \in N$. Clearly, if N and M are manifolds with $dimN = n$ and $dimM = m$, the injectivity and surjectivity of the differential $F_{*,p} : T_pN \to T_pM$ imply $n \leq m$ and $n \geq m$ respectively. In particular, an immersion need not be one-one and a submersion need not be onto. For example, if U is an open subset of a manifold M, then the inclusion $i : U \to M$ is both an immersion and a submersion, but clearly i is not onto.

Later, we will undertake more in-depth analysis of immersions and submersions. According to immersion and submersion theorems to be proven later, every immersion is locally an inclusion and every submersion is locally a projection.

1.5.5 Rank, and Critical and Regular points

Consider a smooth map $F : N \to M$ of manifolds. Its *rank* at a point p in N, denoted by rk $F(p)$, is defined to be the rank of the differential $F_{*,p} : T_pN \to T_pM$. Relative to the coordinate neighbourhoods $(U, x^1, \ldots, x^n)$ at p and $(V, y^1, \ldots, y^m)$ at $F(p)$, the differential is represented by the Jacobian matrix $[\frac{\partial F^i}{\partial x^j}(p)]$, so,

$$\text{rk } F(p) = \text{rk } [\tfrac{\partial F^i}{\partial x^j}(p)].$$

Since, the differential of a map is independent of coordinate charts, so is the rank of a Jacobian matrix.

Now, we define the critical and regular points of a function defined between manifolds.

Definition 1.5.2. A point p in N is a ***critical point*** of F if the differential

$$F_{*,p} : T_pN \to T_pM$$

fails to be surjective.

Definition 1.5.3. A point $c \in M$ is a ***critical value*** if and only if *some* point in the preimage $F^{-1}(c)$ is a critical point.

Definition 1.5.4. A point $p \in N$ is said to be a ***regular point*** of F if the differential $F_{*,p}$ is surjective. In other words, p is a regular point of the map F if and only if F is a submersion at p.

Definition 1.5.5. A point c in the image of F is a ***regular value*** if and only if *every* point in the preimage $F^{-1}(c)$ is a regular point.

Remark 1.5.3. A point in M is a critical value if it is the image of a critical point; otherwise it is a regular value. So, a point not in the image of F automatically becomes a regular value because it is not in the image of a critical point.

1.6 The Tangent Bundle

In this section, we introduce the notion of tangent bundles, which is a specific example of vector bundles over a certain topological space, which in our case is the manifold M. A smooth vector bundle over a smooth manifold M is a smoothly varying family of vector spaces, parametrized by M. One such canonical choice of smooth vector bundles over a smooth manifold M is the collection of tangent spaces to that manifold M, called the *tangent bundle*. The differential of a smooth map between two smooth manifolds induces a bundle map between the corresponding tangent bundles. Thus, the tangent bundle construction is a functor from the category of smooth manifolds to the category of vector bundles.

Since, a tangent bundle is canonically associated to a manifold, invariants of the tangent bundle will give rise to invariants of the manifold. Chern-Weil theory of characteristic classes uses differential geometry for construction of invariants of vector bundles. In particular, applied to tangent bundles, Euler characteristic is one of the numerical diffeomorphism invariants for a manifold.

Later, we will discuss about smooth sections of a vector bundle, wherein a section of a vector bundle $\pi : E \to M$ is a map that maps each point $x \in M$ into the fiber of the bundle $\pi^{-1}\{x\}$. As we shall see, vector fields and differential forms on a manifold are sections of vector bundles over the manifold.

1.6.1 The Topology of the Tangent Bundle

Let M be a smooth manifold. We know, for each point p in M, the tangent space T_pM is the vector space of all one point derivations of the algebra of germs of C^∞ functions at p. The *tangent bundle* at M is the disjoint union of all the tangent spaces at M, that is,

$$TM = \bigsqcup_{p \in M} T_pM.$$

We now give a topology on TM.

Let U be coordinate neighbourhood with coordinates $(x_1, \dots, x_n)$ and $\phi : U \to \phi(U) \subseteq \mathbb{R}^n$.

Let, v_p[4] $\in TU$. Then, $v_p = \sum_{i=1}^{n} c_i \frac{\partial}{\partial x_i}\Big|_p$, where c_i are constants depending on p and v_p.

[4]We switch between the notations v_p and X_p of tangent vectors as per convenience.

Let us define,

$$\begin{aligned}\tilde{\phi}: TU &\to \phi(U) \times \mathbb{R}^n \\ v_p &\mapsto (\phi(p), c_1, \dots, c_n).\end{aligned}$$

Since, ϕ is a homeomorphism and by definition of the coordinate representation of the tangent vectors, it is clear that $\tilde{\phi}$ is a bijection.

Now, the topology on TU is given as: A set $A \subseteq TU$ is open iff $\tilde{\phi}(A)$ is open in $\phi(U) \times \mathbb{R}^n \subseteq \mathbb{R}^{2n}$. Considering the smooth atlas $\{(U_\alpha, \phi_\alpha)\}_{\alpha \in \Lambda}$ of M, we get the following isomorphism:

$$\{TU_\alpha\}_\alpha \cong \phi_\alpha(U_\alpha) \times \mathbb{R}^n.$$

We define the following set :

$\mathcal{B} = \{A \subseteq TU$ is open| U is a coordinate neighbourhood$\}$.

and show that it's a basis for some topology on TM. And we give TM the topology generated by the basis $\mathcal{B}$.

To show that $\mathcal{B}$ is a basis, we need to verify the following two conditions:

$i)$ $\cup_{A \in \mathcal{B}} A = TM$

$ii)$ if$A \subset TU$ and$B \subset TV$ are open, then,$A \cap B$ is open in$T(U \cap V)$.

For (i), we have $\cup_{\alpha \in \Lambda} TU_\alpha = TM$. (ii) is also easy to see just by considering the subspace topology on $T(U \cap V)$. Since, $A \cap T(U \cap V)$ is open in $T(U \cap V)$, and $B \cap T(U \cap V)$ is open in $T(U \cap V)$, it easily follows that $A \cap B \cap T(U \cap V)$ is open, hence, $A \cap B$ is open in $T(U \cap V)$.

It is remarkable to note that if $V \subset U$ is open, then the topology on TV that we get by pulling back the map $\tilde{\phi} : TV \to \phi(V) \times \mathbb{R}^n$ coincides with the subspace topology of TV inherited from TU.

Lemma 1.6.1. *TM is second countable.*

Proof. We know that every manifold can be covered with a countable collection of coordinate neighbourhoods. Let $\{(\phi_i, U_i)\}_{i=1}^\infty$ be a collection of coordinate charts such that $\{U_i\}_{i=1}^\infty$ forms a countable basis for M consisting of coordinate open sets. Since, TU_i is homeomorphic to the open subset $\phi(U_i) \times \mathbb{R}^n$ of $\mathbb{R}^{2n}$ and any subset of an Euclidean space is second countable, it follows that TU_i is second countable. So, for each i, we can choose a countable basis $\{B_{i,j}\}_{j=1}^\infty$ for TU_i. Then, $\{B_{i,j}\}_{i,j=1}^\infty$ is a countable basis for the tangent bundle TM. □

It is also very easy to see that TM is Hausdorff. Now, the only thing left is to give our topological space TM a smooth manifold structure.

$\{(TU_\alpha, \tilde{\phi_\alpha})\}_{\alpha \in \Lambda}$ is an atlas for TM where $\{(U_\alpha, \phi_\alpha)\}_{\alpha \in \Lambda}$ is an atlas for M. Hence, TM is a topological manifold of dimension 2n. We show that TM with this atlas is a smooth manifold. We have $TU_\alpha \cap TU_\beta \neq \phi$ iff $U_\alpha \cap U_\beta \neq \phi$. For simplification, we call $U_\alpha \cap U_\beta$ as $U_{\alpha\beta}$. We show that the following transition map is smooth:

$$\tilde{\phi_\alpha} \circ \tilde{\phi_\beta}^{-1} : \tilde{\phi}_\beta(TU_{\alpha\beta}) \to \tilde{\phi}_\alpha(TU_{\alpha\beta})$$

Consider a point $p \in U_{\alpha\beta}$ and $v_p \in T(U_{\alpha\beta})$.
Then,

$$v_p = \sum_i c_i^\alpha \frac{\partial}{\partial x_i^\alpha}\Big|_p = \sum_i c_i^\beta \frac{\partial}{\partial x_i^\beta}\Big|_p.$$

Hence, we have,

$$\begin{aligned}\tilde{\phi}_\alpha \circ \tilde{\phi}_\beta^{-1}(v_1, \dots, v_n, c_1^\beta, \dots, c_n^\beta) &= \tilde{\phi}_\alpha(\sum_i c_i^\beta \frac{\partial}{\partial x_i^\beta}|_p) \\ &= (\phi_\alpha \circ \phi_\beta^{-1}(v_1, \dots, v_n), c_1^\alpha, \dots, c_n^\alpha).\end{aligned}$$

which shows that TM is a smooth manifold of dimension $2n$.

Tangent bundles being a special case of vector bundles, also similarly comes with a natural projection map:

$$\begin{aligned}\pi : TM &\to M \\ (v_1, \dots, v_n, c_1, \dots c_n) &\mapsto (v_1, \dots, v_n)\end{aligned}$$

1.7 Vector Fields

A vector field on a manifold M is a section of the tangent bundle TM of M, as to each point $p \in M$, a vector field X on M assigns a tangent vector X_p in T_pM. So, we define a vector field as smooth if it is smooth as a section of the tangent bundle.

The notion of vector fields is not only bound to the concept of a manifold as it is quite intrinsic to note that every smooth vector field may be viewed locally as the velocity vector field of a fluid flow. The path traced out by a point under this flow is called the *integral curve* of the vector field. We use the methods of ODE , that is, solve a system of first order ordinary differential equation to find the equation of such an integral curve. Thus, the theory of ODE guarantees the existence of integral curves.

The set of all C^∞ vector fields on a manifold M clearly has the structure of a vector space and we denote it by $\mathfrak{X}(M)$. Unlike tangent vectors, smooth vector fields do not push forward under smooth maps. A smooth bijection is enough to define a unique rough (that is not necessarily smooth) vector field but it is only a diffeomorphism between two smooth manifolds , say, $F : M \to N$ that suffices the push forward of a smooth vector field, say, $X : M \to TM$ to a smooth vector field $F_*X : N \to TN$.

For example, if we consider the following smooth (clearly not a diffeomorphism) global chart on $\mathbb{R}$:

$$\begin{aligned}F : \mathbb{R} &\longrightarrow \mathbb{R} \\ x &\longmapsto x^3.\end{aligned}$$

and the constant vector field $X_p = \frac{\partial}{\partial x}|_p$, where p is a point in M, then , F_*X is defined by:

$$\begin{aligned}(F_*X)_p &= dF_{F^{-1}(p)}(X_{F^{-1}(p)}) \\ &= dF_{p^{1/3}}(\frac{\partial}{\partial x}|_{p^{1/3}}) \\ &= 3p^{2/3}(\frac{\partial}{\partial x}|_p)\end{aligned}$$

which is clearly not a smooth vector field.

1.7.1 Smoothness of a vector field

We have already defined a vector field $X : M \to TM$ to be smooth if it is smooth a section of the tangent bundle $\pi : TM \to M$. In this and the following subsections, we define the smoothness of a vector field in two ways: in terms of coefficient of the vector field on a chart on the manifold M and in terms smooth functions defined on the manifold M.

Let us consider a coordinate chart, say, $(U, \phi) = (U, x^1, x^2, \ldots, x^n)$ on the manifold M. Let, $p \in U$ be a point. The value of the vector field X at the point p is given by the linear combination:

$$X_p = \sum a^i(p) \frac{\partial}{\partial x^i}|_p.$$

It is clear that as p varies in U.

Now, as we saw in the construction of tangent bundles, every such chart (U, ϕ) on M induces a chart $(TU, \tilde{\phi})$ on TM, where:

$$(TU, \tilde{\phi}) = (TU, \tilde{x^1}, \ldots, \tilde{x^n}, c^1, \ldots, c^n)^5$$

$\tilde{x^i} = x^i \circ \pi$, and the c^i's are defined as:

$$v_p = \sum_i c^i(v_p) \left.\frac{\partial}{\partial x^i}\right|_p, \text{ where } v_p \in T_pM.$$

Comparing the coefficients in :

$$X_p = \sum a^i(p) \left.\frac{\partial}{\partial x^i}\right|_p = \sum_i c^i(X_p) \left.\frac{\partial}{\partial x^i}\right|_p$$

we get $a^i = c^i \circ X$ as functions on U. Now, c^i being coordinates are smooth functions on TU. So, if X is a smooth vector field and $(U, x^1, \ldots, x^n)$ is chart on the manifold M containing the point p, then, the coefficients a^i of the the vector field X are smooth functions on U relative to the frame $\left.\frac{\partial}{\partial x^i}\right|$. It is interesting to note that the converse of this is also true.

To conclude, if we are given a manifold M and we consider a chart $(U, x^1, \ldots, x^n)$ around a point p in M, then if X is a vector field flowing through the manifold M, which also assigns a tangent vector v_p at the point p, then using the smoothness of the coordinate functions c^i as above, we get that the coordinates a^i are smooth iff the vector field X is smooth.

We are now ready to characterize the smoothness of a vector fields, the first of which is given by the following proposition:

Proposition 1.7.1. ***(Smoothness of a vector field in terms of coefficients).*** *Let X be a vector field on the manifold M. The following are equivalent :*

1. *The vector field X is smooth on M.*

2. *The manifold M has an atlas such that corresponding to any chart $(U, x^1, \ldots, x^n)$ of that atlas, the coefficients a^i of the vector field $X = \sum a^i \left.\frac{\partial}{\partial x^i}\right|$ relative to the frame $\left.\frac{\partial}{\partial x^i}\right|$ are all smooth.*

[5] Indexed variables are just notations. So we oscillate between superscripts and subscripts.

The second part of the above proposition considers the existence of such an atlas. But it is not difficult to note that if X is a smooth vector field o the manifold M, then on any chart of the manifold M, this vector field will be smooth, that is, it hold for any atlas defined on the manifold M. (By any atlas we obviously mean a smooth atlas as we are always dealing with a smooth manifold M).

As we have stated before, we can also characterize smooth vector fields on the manifold M in terms of smooth functions defined on the manifold. It is given by the following proposition :

Proposition 1.7.2. *A vector field X on M is smooth iff for every smooth function f defined on M, the function Xf is smooth on M.*

Proof. Will write when I get time. □

We not only can think smooth vector fields on a manifold M as smooth sections of the tangent bundle $\pi : TM \to M$, but also as derivations on algebra of C^∞ functions on the manifold M. And in fact, these two definitions are equivalent!

By proposition 1.7.2, we can view a smooth vector field $X : M \to TM$ as a linear operator $X : C^\infty(M) \to C^\infty(M)$ which is also a derivation, since, for all $f, g \in C^\infty(M)$, we have:

$$X(fg) = (Xf)g + f(Xg).$$

As we find in the case of smooth functions in lemma 1.4.5, there is a similar analogue of C^∞ extension of vector fields from a possibly smaller neighbourhood of a manifold to the entire manifold. It is given by the following proposition :

Proposition 1.7.3. *(C^∞ **extension of vector fields**). If X is a C^∞ vector field defined on a neighbourhood U of a point p in a manifold M, then there exists a C^∞ vector field $\tilde{X}$ which agrees with X on a possibly smaller neighbourhood of p.*

Proof. The proof goes similar to the proof of lemma 1.4.5 by considering the following function:

$$\tilde{X}(q) = \begin{cases} \rho(q)X(q) & \text{if q} \in U \\ 0 & \text{if q} \notin U \end{cases}$$

where $\rho : M \to \mathbb{R}$ is a smooth bump function supported in a neighbourhood U of p which is identically equal to 1 in a neighbourhood $V(\subset U)$ of p. □

1.8 Submanifolds

Most manifolds can be naturally described as *submanifolds* of something simpler. Now, to define the notion of a *submanifold*, we motivate ourselves from our well-known ***Implicit Function Theorem*** which states : Let $f : \mathbb{R}^n \to \mathbb{R}$ be a continuously differentiable (C^∞ *or smooth*) function. Let us consider the zero set, say S, of f, where:

$$S := \{\mathbf{x} = (x = (x_1, \dots, x_{n-1}), y) \in \mathbb{R}^n : f(\mathbf{x}) = 0\}$$

For any point $(a, b) \in S$, where $a = (a_1, \dots, a_{n-1})$, such that $\frac{\partial f}{\partial y}(a, b) \neq 0$, $\exists$ an open *nbd* $U \subseteq \mathbb{R}^{n-1}$ of a and $V \subset \mathbb{R}$ of b and a smooth function $h : U \to V$ such that $h(a) = b$ and $f(x, h(x)) = 0, \forall x \in U$, and taking $W = U \times V$, we have,

$$W \cap S = \{(x, h(x)) : x \in \mathbb{R}^{n-1}\},$$

i.e. the zero set of the function f can be locally represented as the graph of a smooth function.

So, in general, if $f : \mathbb{R}^n \to \mathbb{R}$ is a smooth function such that the zero set, say S, of f contains a regular point (or if 0 is a regular value of f), then S can be locally represented as the graph of a smooth function defined around an open *nbd* of that point (or any point of S).

Our main motive to bring in Implicit Function Theorem is to describe level sets of smooth functions as submanifolds. Now, to venture into the quest of understanding what it actually means, which we will later see in *Regular Level Set Theorem*, let us first give the general definition of a *submanifold* to make things clear.

Definition 1.8.1. Let M be a smooth manifold of dimension m, and N be its subset. Then, N is called a ***smooth n-dimensional submanifold*** *of* M if for every $p \in N$, there is a smooth chart (U, ϕ) in M such that $p \in U$ and $\phi(N \cap U) = \mathbb{R}^n \cap \phi(U)$, where $\mathbb{R}^n$ is embedded into $\mathbb{R}^m$ as the subspace $\{x_{n+1} = 0, \ldots, x_m = 0\}$.

We now use Implicit Function Theorem to give the following proposition:

Proposition 1.8.1. *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a smooth function and $N = \{x \in \mathbb{R}^n : f(x) = 0\}$, the zero set of f. Suppose the gradient of f, i.e. the vector of its partial derivatives does not vanish on N, or if 0 is a regular value of f, then N is a smooth submanifold of dimension $(n-1)$ (or of codimension 1).*

Proof. Let $p = (a_1, \ldots, a_n) \in N$. Since, the gradient of f does not vanish at p, there is $i \in \{1, 2, \ldots, n\}$ such that $\frac{\partial f}{\partial x_i} \neq 0$. Without loss of generality, we first assume that $i = n$. Then, by Implicit Function Theorem, there is a neighbourhood $U_a \subseteq \mathbb{R}^{n-1}$ of $(a_1, \ldots, a_{n-1})$ and U_b of a_n and a smooth function $h : U_a \to U_b$ such that $f(x_1, \ldots, x_{n-1}, h(x_1, \ldots, x_{n-1})) = 0$, $\forall (x_1, \ldots, x_{n-1}) \in U_a$. Here, $x_n = h(x_1, \ldots, x_{n-1})$. Let, $W = U_a \times U_b$. Then,

$$\begin{gathered} \phi_W : W \cap N \to \mathbb{R}^{n-1} \\ (x_1, \ldots, x_n) \mapsto (x_1, \ldots, x_{n-1}) \end{gathered}$$

is a coordinate chart.
Similarly, WLOG if $i = n-1$, then, $x_{n-1} = h'(x_1, \ldots, x_{n-2}, x_n)$ where $h' : U'_{a'} \to U'_{b'}$ is the corresponding smooth function we obtain by IFT, and $U'_{a'}$ is an open *nbd* of $a' = (a_1, \ldots, a_{n-2}, a_n)$ and $U'_{b'}$ is an open *nbd* of a_{n-1}, $W' = U_{a'} \times U_{b'}$, then, the coordinate chart is given by:

$$\begin{gathered} \psi_{W'} : W' \cap N \to \mathbb{R}^{n-1} \\ (x_1, \ldots, x_n) \mapsto (x_1, \ldots, x_{n-2}, x_n) \end{gathered}$$

Then, the transition functions are given by:

$$\begin{gathered} \psi_{W'} \circ \phi_W^{-1} : \phi_W(W \cap W' \cap N) \to \psi_{W'}(W \cap W' \cap N) \\ (x_1, \ldots, x_{n-1}) \mapsto (x_1, \ldots, x_{n-2}, h(x_1, \ldots, x_{n-1})), \text{ and,} \end{gathered}$$

$$\begin{gathered} \phi_W \circ \psi_{W'}^{-1} : \psi_{W'}(W \cap W' \cap N) \to \phi_W(W \cap W' \cap N) \\ (x_1, \ldots, x_{n-2}, x_n) \mapsto (x_1, \ldots, x_{n-2}, h'(x_1, \ldots, x_{n-2}, x_n)) \end{gathered}$$

both of which are clearly smooth. Since, $p \in N$ is arbitrary, N is a smooth submanifold of codimension 1. □

Remark 1.8.1. In the above proposition,

$$W \cap N = \{(x, h(x)) : x \in \mathbb{R}^{n-1}\}$$

So, the zero set of f locally looks like the graph of a smooth function.

Remark 1.8.2. Let, $\mathcal{G}(f)$ be the graph of a smooth function from $\mathbb{R}^n$ to $\mathbb{R}$. Then, $\mathcal{G}(f) = \{(x, f(x)) : x \in \mathbb{R}^n\}$. By taking projection, say ϕ, onto the first component of the graph, we obtain our required chart (U_x, ϕ), where U_x is diffeomorphic with an open subset of $\mathbb{R}^n$. Since, x is arbitrary, $\mathcal{G}(f)$ is a submanifold of $\mathbb{R}^{n+1}$ of dimension n.

Corollary 1.8.1. *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a smooth function and $N_c = \{x \in \mathbb{R}^n : f(x) = c\}$, the level set of f at c, where $c \in \mathbb{R}$. Suppose the gradient of f, i.e. the vector of its partial derivatives does not vanish on N_c, then N is a smooth submanifold of codimension 1.*

Proof. This follows from the above proposition taking the function to be $f - c$. $\square$

Now, we can generalize the above corollary from the case of $\mathbb{R}^n$ and $\mathbb{R}$ to the case of arbitrary manifolds N and M of dimension n and m respectively, which gives our popularly known theorem, *the regular level set theorem*. For this, we use *Inverse Function Theorem for manifolds*. We are all familiar with the inverse function theorem on $\mathbb{R}^n$ from our calculus courses, but we can further generalize this to arbitrary manifolds: Let $F : N \to M$ be a C^∞ map between manifolds of the same dimension and $p \in N$. Suppose for some charts $(U, \phi) = (U, x^1, \dots, x^n)$ about p in N and $(V, \psi) = (V, y^1, \dots, y^n)$ about $F(p)$ in M, $F(U) \subset V$. Set $F^i = y^i \circ F$. Then F is locally invertible about p if and only if its Jacobian determinant $det[\frac{\partial F^i}{\partial x^j}(p)]$ is non-zero.

Theorem 1.8.1. ***(Regular Level Set Theorem)*** *Let $F : N \to M$ be a C^∞ map on manifolds, with $dimN = n$ and $dimM = m$. Then a non-empty regular level set $S = F^{-1}(c)$, where $c \in M$, is a regular submanifold of N of dimension $n - m$.*

Proof. We leave the proof for later.

$\square$

1.9 Multilinear Algebra

Manifolds are higher-dimensional analogues of curves and surfaces. As we noted earlier the importance of linearization principle in manifold theory, here in this section we will try to give a broad view of it. Since, every higher dimensional manifolds are not linear spaces, we approximate our manifolds locally by its tangent space at a point which is a linear object. Instead of working with tangent vectors, it is more useful to adopt a dual point of view and work with linear functions on a tangent space, as being functions, they can be added, multiplied, scalar multiplied and composed with other functions. After that, it is but a small step to consider functions of several arguments, linear in each argument. These are called *multilinear functions* on a vector space.

1.9.1 Multilinear functions

A function $f : V^k$ *(k copies of a real vector space)* $\to \mathbb{R}$ is *k-linear* if it is linear in each of its k arguments:

$$f(\dots, av + bw, \dots) = af(\dots, v, \dots) + bf(\dots, w, \dots) \text{ for all } a, b \in \mathbb{R} \text{ and } v, w \in V.$$

A linear function on V is also called a $k - tensor$ on V. We denote the vector space of all *k-tensors* on V by $L_k(V)$.

Example 1.9.1. *The determinant $f(v_1, \dots, v_n) = det[v_1, \dots, v_n]$ viewed as a function of n column vectors $v_1, \dots, v_n$ in $\mathbb{R}^n$, is n-linear.*

Definition 1.9.1. A k-linear function $f : V^k \to \mathbb{R}$ is *symmetric* if

$$f(v_{\sigma(1)},\dots,v_{\sigma(k)}) = f(v_1,\dots,v_k)$$

for all permutations $\sigma \in S_k$; it is *alternating* if

$$f(v_{\sigma(1)},\dots,v_{\sigma(k)}) = (\text{sgn } \sigma) f(v_1,\dots,v_k)$$

for all $\sigma \in S_k$.

For example, the dot product $f(v,w) = v \cdot w$ on $\mathbb{R}^n$ is symmetric, whereas, the above defined determinant function $f(v_1,\dots,v_n) = det[v_1,\dots,v_n]$, of its n column vectors, is alternating.

We deal especially with the space of all alternating k-linear functions and denote it by $A_k(V)$ $(k > 0)$. These are also called *alternating k-tensors or k-covectors.*

Given any k-linear function on a vector space V, there is a way to construct symmetric k-linear functions Sf and alternating k-linear functions Af from it, wherein,

$$Sf = \sum_{\sigma\in S_k} \sigma f.$$
$$Af = \sum_{\sigma\in S_k} (\text{sgn } \sigma)\sigma f.$$

Now we define tensor products which are heavily used in differential forms, as we will see soon.

1.9.2 The Tensor Product

Let f be a k-linear function and g be a l-linear function on a vector space V. Their ***tensor product*** is a $(k+l)$-linear function $f \otimes g$ defined by

$$(f \otimes g)(v_1,\dots,v_{k+l}) = f(v_1,\dots,v_k)g(v_{k+1},\dots,v_{k+l})$$

Let us see the following example of bilinear maps expressed in terms of tensor product:

Let $e_1,\dots,e_n$ and $\alpha^1,\dots,\alpha^n$ be the basis for a vector space V and its dual space $V^\vee$ respectively. And, let $\langle,\rangle : V \times V \to \mathbb{R}$ be a bilinear map on V. Then, we have $v = \Sigma v^i e_i$, $w = \Sigma w^i e_i$ and $\alpha^i(v) = v^i$ and $\alpha(w) = w^i$ and setting $g_{ij} = \langle e_i, e_j\rangle$, we get:

$$\langle v, w\rangle = \sum v^i w^j \langle e_i, e_j\rangle = \sum \alpha^i(v)\alpha^j(w) g_{ij} = \sum g_{ij}(\alpha^i \otimes \alpha^j)(v,w).$$

Hence, $\langle,\rangle = \sum g_{ij}\alpha^i \otimes \alpha^j$. This notation is often used to described inner products on a vector space.

Now, let us define what a wedge product is.

1.9.3 The Wedge Product

If f and g are two alternating multilinear functions, then it is obvious to ask for a product which is also alternating. That's where we are motivated to define the ***wedge product***, also called ***exterior product***: for $f \in A_k(V)$ and $g \in A_l(V)$,

$$f \wedge g = \frac{1}{k!l!} A(f \otimes g); \tag{1.1}$$

or explicitly,

$$(f \wedge g)(v_1, \ldots, v_{k+l}) = \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} (\text{sgn } \sigma) f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}) g(v_{\sigma(k+1)}, \ldots, v_{\sigma(k+l)}) \tag{1.2}$$

By definition of alternating multilinear functions, $f \wedge g$ is clearly alternating.

When $k = 0, f \in A_0(V)$ is simply a constant c. In this case, $c \wedge g$ is just a scalar multiplication, and thus $c \wedge g = cg$ for $c \in \mathbb{R}$ and $g \in A_l(V)$.

1.9.4 Properties of Tensor and Wedge products

1. If f, g, h are multilinear functions on V, then

    (a) The tensor product is associative: $(f \otimes g) \otimes h = f \otimes (g \otimes h)$.

    (b) The wedge product is associative: $(f \wedge g) \wedge h = f \wedge (g \wedge h)$.

2. The wedge product is anti-commutative: if $f \in A_k(V)$ and $g \in A_l(V)$, then,

$$f \wedge g = (-1)^{kl} g \wedge f.$$

3. Suppose f is a k-linear function and g is a l-linear function on a vector space V. Then

    (a) $A(A(f) \otimes g) = k!A(f \otimes g)$,and

    (b) $A(f \otimes A(g)) = l!A(f \otimes g)$.

Corollary 1.9.1. *If f is a multicovector of odd degree on V, then $f \wedge f = 0$.*

Proof. Let, k be the degree of f . By 5.4.2, we have $f \wedge f = (-1)^{k^2} f \wedge f$. Since, k is odd, $f \wedge f = 0$. □

Corollary 1.9.2. *Let $f_i \in A_{d_i}(V)$, then*

$$f_1 \wedge \cdots \wedge f_r = \frac{1}{(d_1)! \ldots (d_r)!} A(f_1 \otimes \cdots \otimes f_r). \tag{1.3}$$

In particular, we have the following corollary. We use the notation $[b^i_j]$ to denote the matrix whose (i, j)-entry is b^i_j.

Proposition 1.9.1. ***(Wedge product of 1-covectors).*** *If $\alpha^1, \ldots, \alpha^k$ are linear functions on a vector space V and $v_1, \ldots, v_k \in V$, then*

$$(\alpha^1 \wedge \cdots \wedge \alpha^k) = det[\alpha^i(v_j)].$$

Proof. By Corollary 1.9.2,

$$\begin{aligned}
(\alpha^1 \wedge \cdots \wedge \alpha^k)(v_1, \ldots, v_k) &= A(\alpha^1 \otimes \cdots \otimes \alpha^k)(v_1, \ldots, v_k) \\
&= \sum_{\sigma \in S_k} (\text{sgn } \sigma) \alpha^1(v_{\sigma(1)}) \ldots \alpha^k(v_{\sigma(k)}) \\
&= \det[\alpha^i(v_j)].
\end{aligned}$$

□

1.9.5 Graded algebras

Graded algebras are much used in commutative algebra and algebraic geometry, homological algebra, and algebraic topology. Also, in the field of mathematical physics, graded lie algebras are of much importance, for example, they are used in describing some quantum field theoretic operators.

Graded algebras are associative algebras such that each elements of this algebra are labelled by a monoid and any multiplication action in this algebra is reflected in the multiplication of the labelling group. For example, a ***graded ring*** is a ring R equipped with a decomposition of its underlying abelian group G as a direct sum $R = \bigoplus_{g \in G} R_g$ such that the product map takes $R_g \times R_{g'}$ to $R_{gg'}$.

There is a notion of graded k-associative algebra over any commutative ring k, where, in particular, a ***graded algebra*** A over a field K can be written as a direct sum $A = \bigoplus_{k=0}^{\infty} A^k$, where A^k are vector spaces over K such that the multiplication map sends $A^k \times A^l$ to A^{k+l}. The notation $A = \bigoplus_{k=0}^{\infty} A^k$ means that every non zero element of A is uniquely a *finite* sum

$$a = a_{i_1} + \cdots + a_{i_m},$$

where $a_{i_j} \neq 0 \in A^{i_j}$. Clearly, here the grading is done by non-negative integers.

A graded algebra $A = \bigoplus_{k=0}^{\infty} A^k$ is said to be *anti-commutative or graded commutative* if for all $a \in A^k$ and $b \in A^l$,

$$ab = (-1)^{kl} ba.$$

A *homomorphism of graded algebras* is an algebra homomorphism that preserves the degree.

After few subsections, we will introduce the notion of smooth k-forms which is an example of *differential graded algebras*, which is a graded algebra A equipped with a derivation $d : A \to A$ of degree +1 (or -1 depending on conventions) such that $d \circ d = 0$. This is same as a monoid in the category of chain complexes.

Example 1.9.2. *The polynomial algebra $A = \mathbb{R}[x, y]$ is graded by degree; A^k consists of all homogeneous polynomials of total degree k in the variables x and y.*

Now, let suppose, V is a finite dimensional vector space, say of dimension n, then we define

$$A_*(V) = \bigoplus_{k=0}^{\infty} A_k(V) = \bigoplus_{k=0}^{n} A_k(V).$$

By the definition of wedge product of multi-covectors and from 5.4.2, it is clear that $A_*(V)$ becomes an anti-commutative graded algebra, called the *exterior algebra* or the *Grassmann algebra* of multi-covectors on the vector space V.

1.9.6 A basis for the space of k-covectors

Let $e_1, \ldots, e_n$ be a basis for a real vector space V, and, let $\alpha^1, \alpha^2, \ldots, \alpha^n$ be the dual basis for $V^\vee$. Let us set the multi-index notation

$$I = (i_1, \ldots, i_k)$$

and write e_I for $(e_{i_1}, \ldots, e_{i_k})$ and α^I for $\alpha^{i_1} \wedge \cdots \wedge \alpha^{i_k}$. It is clear that if f is alternating, then it suffices to find its values on k-tuples $(e_{i_1}, \ldots, e_{i_k})$, where $1 \leq i_1 < \cdots < i_k \leq n$, that is, it suffices to consider e_I with I in strictly ascending order.

Lemma 1.9.1. *Let $(e_{i_1}, \ldots, e_{i_n})$ be a basis for a vector space V and $\alpha^1, \alpha^2, \ldots, \alpha^n$ be its dual basis in $V^\vee$. If $I = (1 \leq i_1 < \cdots < i_k \leq n)$ and $J = (1 \leq j_1 < \cdots < j_k \leq n)$ are strictly ascending multi-indices of length k, then*

$$\alpha^I(e_J) = \delta^I_J = \begin{cases} 1 & \textit{for } I = J \\ 0 & \textit{for } I \neq J \end{cases}$$

Proof. This easily follows from Proposition 1.9.1 by considering the cases $I = J$ and $I \neq J$, wherein the first case gives the identity matrix, hence has determinant 1. And, in the second case, considering the tuples I and J term by term, we find that one of the rows of our matrix become completely zero, thus giving zero determinant. Hence, the result follows. □

Next, we give the following proposition for which we did all our above described constructions.

Proposition 1.9.2. *The alternating k-linear functions α^I, $I = (1 \leq i_1 < \cdots < i_k \leq n)$, form a basis for the space $A_k(V)$ of alternating k-linear functions on V.*

Proof. First, we prove the linear independence. Suppose $\sum c_I \alpha^I = 0, c_I \in \mathbb{R}$, and I runs over all strictly ascending multi-indices of length k. Evaluating both sides at e_J, $J = (j_1 < \cdots < j_k)$, we have by lemma 1.9.1,

$$0 = \sum_I c_I \alpha^I(e_J) = \sum_I c_I \delta^I_J = c_J,$$

since over all strictly increasing multi-indices I of length k, there is only one equal to 0. This proves the linear independence of α^I's.

Now, to show that $\alpha^I \in$ span $A_k(V)$, let $f \in A_k(V)$. We claim that

$$f = \sum f(e_I)\alpha^I$$

where I runs over all strictly increasing multi-indices of length k. let, $g = \sum f(e_I)\alpha^I$. By, k-linearity and the alternating property, if the two k-covectors agree on all e_J, where $J = (j_1 < \cdots < j_k)$, then they are equal. But,

$$g(e_J) = \sum f(e_I)\alpha^I(e_J) = \sum f(e_I)\delta^I_J = f(e_J).$$

Therefore, $f = g = \sum f(e_I)\alpha^I$. □

Corollary 1.9.3. *If the vector space V has dimension n, then the vector space $A_k(V)$ of k-covectors on V has dimension $\binom{n}{k}$.*

Corollary 1.9.4. *If $k > dimV$, then, $A_k(V) = 0$.*

1.10 Differential Forms on $\mathbb{R}^n$

Differential forms has far reaching consequences in geometry, topology and physics. In fact, certain physical concepts such as electricity and magnetism are best formulated in terms of differential forms. In our study, we will see how differential forms extends Grassmann's exterior algebra from the tangent space at a point globally to an entire manifold. Just as a vector field, say,

$$\begin{aligned} \phi : U \subseteq \mathbb{R}^n &\longrightarrow T_pU \simeq T_p\mathbb{R}^n \\ p &\longmapsto X_p \end{aligned}$$

assigns a tangent vector to each point of an open subset $U \subseteq \mathbb{R}^n$, so dually, a differential k-form assigns to each point a k-covector on the tangent space to each point of U, as we shall see soon. The wedge product of differential forms is defined pointwise as the wedge product of multicovectors.

When we are studying differential forms, another interesting concept which we often come across is finding the derivative of these differential forms, which is well defined since differential forms exist on an open set, and not just at a single point. The notion of differentiation of differential forms is fascinating since we define it using standard coordinates of $\mathbb{R}^n$, but the derivative comes out to be independent of these coordinates. We call these derivatives as *exterior derivative.*

In this section, we study the simplest case, that of differential forms on an open subset of $\mathbb{R}^n$.

1.10.1 Differential 1-Forms

As we define the dual space $V^\vee$ of a vector space V, we similarly define the dual space $(T_p\mathbb{R}^n)^\vee$ of the tangent space $T_p\mathbb{R}^n$, denote it by $T_p^*\mathbb{R}^n$, and we call it the cotangent space to $\mathbb{R}^n$ at p. Thus an element of $T_p^*\mathbb{R}^n$ is a covector or a linear functional on the tangent space $T_p\mathbb{R}^n$. In parallel to the vector field ϕ defined above, a *covector field* or a *differential 1-form* on an open subset U of $\mathbb{R}^n$ is a function ω that assigns to each point p in U a covector $\omega_p \in T_p^*\mathbb{R}^n$,

$$\begin{aligned} \omega : U \to \bigcup_{p \in U} T_p^*\mathbb{R}^n, \\ p \mapsto \omega_p \in T_p^*\mathbb{R}^n. \end{aligned}$$

It is important to note that the union $\bigcup_{p\in U} T_p^*\mathbb{R}^n$ is a disjoint union of the sets $T_p^*\mathbb{R}^n$. It is because the tangent space itself being the set of all point derivations of C_p^∞, the algebra of germs of C^∞ real valued functions at the point $p \in U \subseteq \mathbb{R}^n$, and these algebras being pairwise disjoint for every two distinct point in U, it follows that the tangent spaces are disjoint for every two distinct points in U. Hence, their duals, that is, the cotangent spaces $T_p^*\mathbb{R}^n$ are also disjoint.

We call a differential 1-form a 1-form for short.

Similarly, we can define the notion of a differential k-form. It is a function ω that assigns to each point p in U, a k-vector $\omega_p \in T_p^*\mathbb{R}^n$.

1.11 Differential forms on a Manifold

Since, now we have an idea what differential forms are, restricting our discussion to $\mathbb{R}^n$ is not interesting when all these notions can be easily generalised to a manifold. So now, further properties of differentials will be introduced in the case of a general manifold and not just on an open subset of $\mathbb{R}^n$.

Definition 1.11.1. If $f \in C^\infty(M)$, its differential is defined to be the 1-form df such that for any $p \in M$ and tangent vector $X_p \in T_pM$,

$$(df)_p X_p = X_p f.$$

In the subsection 1.5.2, we already introduced differential of a smooth map defined on a manifold. It is not any abuse in terminology as comparing both these definitions of a differential, we observe that they are indeed same!

Proposition 1.11.1. *If $f : M \to \mathbb{R}$ is a C^∞ function, then for $p \in M$ and $X_p \in T_pM$, we have*

$$f_*(X_p) = (df)_p(X_p) \left.\frac{d}{dt}\right|_{f(p)}$$

Proof. Since, $f_*(X_p) \in T_f(p)\mathbb{R}$, it follows that there exists $a \in \mathbb{R}$ such that:

$$f_*(X_p) = a \left.\frac{d}{dt}\right|_{f(p)}$$

Now, to find the value of a, we evaluate the above expression both sides at t, to get:

$$a = f_*(X_p)(t) = X_p(t \circ f) = X_p f = (df)_p X_p.$$

□

The above proposition clarifies our doubt of using both of these notations as our differential. Indeed, under the canonical identification of the tangent space $T_{f(p)}\mathbb{R}$ with $\mathbb{R}$ as follows:

$$a \longleftrightarrow a \left.\frac{d}{dt}\right|_{f(p)} ,$$

f_* is same as df.

1.11.1 Local expression of a differential 1-form

As we have seen before, vector field assigns a tangent vector to each point of a manifold whereas, a differential k-form assigns a k-covector or a k-linear functional on the tangent space at every point in an open set of a manifold. Indeed, differential forms are dual to the concept of vector fields. So, it is obvious to talk about the smoothness of differential forms a s we did in the case of vector fields. But, to begin with, we first need to specify a basis for cotangent spaces dual to the basis of tangent spaces.

Lemma 1.11.1. *Suppose $\{(U, \phi, x_1, \ldots, x_n)\}$ is a coordinate neighbourhood in a manifold M and $p \in U$. Then, the dual basis of $\{ \left.\frac{\partial}{\partial x_i}\right| \}_{i=1}^n$ is given by $\{dx_i|_p\}_{i=1}^n$.*

Proof. By definition,

$$(dx_i|_p)\Big(\left.\frac{\partial}{\partial x_j}\right|_p \Big) = \left.\frac{\partial}{\partial x_j}\right|_p x_i = \delta^i_j.$$

□

Suppose, ω is a 1-form, then the restriction of ω to an open subset U containing a point $p \in M$ is given by the linear combination

$$\omega|_p = \sum a_i dx_i.$$

In particular if $f \in C^\infty(M)$, then the restriction of the 1-form df to U is given by

$$df|_p = \sum_{i=1}^{n} a_i(p) dx_i|_p, \forall p \in U.$$

To find the coordinate functions $a_i(p)$, we evaluate both sides at $\frac{\partial}{\partial x_j}\Big|_p$ to get:

$$(df)_p(\frac{\partial}{\partial x_j}\Big|_p) = \sum_{i=1}^{n} a_i(p) dx_i|_p(\frac{\partial}{\partial x_j}\Big|_p)$$

$$\text{i.e. } \frac{\partial f}{\partial x_j}\Big|_p = a_j(p).$$

Hence, we obtain the local expression for df as:

$$df = \sum \frac{\partial f}{\partial x_i}\Big| dx_i.$$

1.11.2 The Cotangent Bundle

The construction of the cotangent bundle is exactly similar to how we constructed the tangent bundle. The underlying set of the cotangent bundle is the disjoint union of cotangent spaces at all points of the manifold, that is,

$$T^*M = \bigsqcup_{p \in M} T_p^*M.$$

Mimicking the case of the tangent bundle, we consider the natural map $\pi : T^*M \to M$, where, $\pi(\alpha) = p$ for each $\alpha \in T_p^*M$ and a chart $(U, \phi, x_1, \ldots, x_n)$ around a point $p \in U$. Now, each $\alpha \in T_p^*M$ can be uniquely represented as:

$$\alpha = \sum_{i=1}^{n} c_i(\alpha) dx_i|_p.$$

This gives rise to get the following bijection:

$$\begin{aligned} \tilde{\phi} : T^*U &\to \phi(U) \times \mathbb{R}^n \\ \alpha &\longmapsto (\phi(p), c_1(\alpha), \ldots, c_n(\alpha)) = (\phi \circ \pi, c_1, \ldots, c_n)(\alpha) \end{aligned}$$

Due to this bijection, we can easily pull back the topology on $\phi(U) \times \mathbb{R}^n$ to give a topology to T^*U considering that any subset of $A \subseteq T^*U$ will be open iff $\tilde{\phi}(A)$ is open in $\phi(U) \times \mathbb{R}^n$. Now, for each coordinate neighbourhood U, the basis $\mathcal{B}_U$ consists of all open subsets of T^*U, and $\mathcal{B}$, which is the union of all such $\mathcal{B}_U$ forms a basis for the topology on T^*M obtained by the pullback of the bijective map, as described earlier. We give T^*M the topology generated by $\mathcal{B}$. We omit the proof of this, as it is exactly similar to what we proved in the case of tangent bundles.

Now, with the maps $\tilde{\phi} = (x_1 \circ \pi, \ldots, x_n \circ \pi, c_1, \ldots, c_n)$ as coordinate maps, the cotangent bundle T^*M becomes a C^∞ manifold (It can be easily observed by computing the transition maps, which are smooth). And, the projection map $\pi : T^*M \to M$ becomes a vector bundle of rank n over the manifold M, and is called the "cotangent bundle".

We can now define a smooth 1-form in terms of the cotangent bundle. A 1-form on M, say, ω is smooth if it is smooth as a section $\omega : M \to T^*M$ of the cotangent bundle $\pi : T^*M \to M$. Since, it is a section, we have that $\pi \circ \omega = \mathbb{1}_M$.

1.11.3 Characterization of C^∞ 1-forms

As we saw in the case of vector fields, differential 1-forms can also be classified as smooth in terms of smooth maps defined on the manifold or whether coefficient functions a_i in its local expression are smooth. The set of all smooth 1-forms defined on the manifold M has the structure of a vector space and are denoted by $\Omega^1(M)$.

Although similar as vector fields, still we state or give proof to some of the propositions:

Proposition 1.11.2. *Let $(U, \phi, x_1, \ldots, x_n)$ be a coordinate chart defined on the manifold M. A 1-form $\omega = \sum a_i dx_i$ is smooth on U if and only if the coefficient function a_i are all smooth.*

Proof. Since the map $\tilde{\phi} : T^*U \to U \times \mathbb{R}^n$ is a diffeomorphism, the 1-form $\omega : U \to T^*U$ is smooth if and only if the composition map $\tilde{\phi} \circ \omega : U \to U \times \mathbb{R}^n$ is smooth, where:

$$\begin{aligned}\tilde{\phi} \circ \omega(p) = \tilde{\phi}(\omega_p) &= (x_1(p), \ldots, x_n(p), c_1(\omega_p), \ldots, c_n(\omega_p)) \\ &= (x_1, \ldots, x_n, a_1(p), \ldots, a_n(p)).\end{aligned}$$

As the coordinate functions $x_1, \ldots, x_n$ are all smooth on U, it is clear that $\tilde{\phi} \circ \omega$ will be smooth on U if and only if all a_i are smooth on U.

□

Proposition 1.7.1 also holds for the case of 1-forms.

Proposition 1.11.3. *If f is a C^∞ function defined on the manifold M, then its differential df is a C^∞ 1-form on M.*

Proof. On any chart $(U, \phi, x_1, \ldots, x_n)$ on the manifold M, the differential df is defined as $df = \sum \left.\frac{\partial f}{\partial x_i}\right| dx_i$. Since, $\left.\frac{\partial f}{\partial x_i}\right|$ are all smooth, it follows that the 1-form df is smooth. □

Since, differential forms are dual to the concept of vector fields, it is quite intuitive for one to think about expressing C^∞ 1-forms in terms of C^∞ vector fields. And, it is indeed possible!

Before we state and proof the proposition which justifies the above, we define a function $\omega(X)$ on M as follows:

$$\omega(X)_p = \omega_p(X)_p \in \mathbb{R}, p \in M.$$

Proposition 1.11.4. ***(Linearity of 1-form over functions).*** *Let ω be 1-form on a manifold M. If f is a function and X is a vector field, then, $\omega(fX) = f\omega(X)$.*

Proof. Since, $\omega(X)$ is defined pointwise, so, for each point $p \in M$, we have:

$$(\omega(fX))_p = \omega_p(f(p)X_p) = f(p)(\omega_p X_p) = (f(\omega X))_p.$$

as ω_p is $\mathbb{R}$-linear in its argument. $\square$

Now, we present the proposition which we actually ought to proof:

Proposition 1.11.5. *A 1-form ω is C^∞ on a manifold M if and only if for every C^∞ vector field X on M, the function $\omega(X)$ is C^∞ on M.*

Proof. Let us consider a chart $(U, \phi, x_1, \ldots, x_n)$ on M. For a C^∞ 1-form $\omega = \sum a_i dx_i$ and a C^∞ vector field $X = \sum b_j \left.\frac{\partial}{\partial x_j}\right|$ on M, we have, by proposition 1.11.4:

$$\omega(X) = (\sum_i a_i dx_i)(\sum_j b_j \left.\frac{\partial}{\partial x_j}\right|) = \sum_{i,j} a_i b_j \delta^i_j = \sum a_i b_i.$$

a C^∞ function on U. Since, U was a arbitrary chart on M, we have that $\omega(X)$ is a C^∞ function on M.

Conversely, suppose ω is a 1-form on M such that for every C^∞ vector field X on M, the function $\omega(X)$ is C^∞ on M. Let us choose a coordinate neighbourhood $(U, \phi, x_1, \ldots, x_n)$ about a point p in M. Then, $\omega = \sum a_i dx_i$ for some functions a_i on U.

Let us fix an integer j, $1 \leq j \leq n$. By C^∞ extension of vector fields, we can extend the vector field $X = \left.\frac{\partial}{\partial x_j}\right|$ defined on the coordinate neighbourhood U to the C^∞ vector field $\tilde{X}$ defined on the entire manifold M that agrees with $\left.\frac{\partial}{\partial x_j}\right|$ on a possibly smaller neighbourhood V_p^j of p contained in U. We now use the smoothness of the function $\omega(\tilde{X})$ to show the smoothness of the coordinate functions a_j on the neighbourhood V_p^j:

$$\omega(\tilde{X}|_{V_p^j}) = \sum a_i dx_i(\left.\frac{\partial}{\partial x_j}\right|) = a_j.$$

In the neighbourhood $V_p := \bigcap_{j=1}^n V_p^j$, all a_j are C^∞. Hence, $\omega = \sum a_i dx_i$ is C^∞ on V_p. Thus, by extension, we get that the 1-form ω is C^∞ on the entire manifold M as for each point p in M, we have found a neighbourhood V_p in which ω is smooth.

$\square$

If $\mathcal{F} := C^\infty(M)$ be the ring of all C^∞ functions over M, then proposition 1.11.5 allows us to define the following map which is both $\mathbb{R}$- linear and $\mathcal{F}$- linear by proposition 1.11.4:

$$\begin{aligned} \omega : \mathfrak{X}(M) &\to \mathcal{F} \\ X &\mapsto X(M). \end{aligned}$$

1.11.4 Pullback of 1-forms

As we have seen, the tangent functor T is a covariant functor from the category of smooth manifolds to the category of smooth vector bundles. To each smooth manifold M, it assigns the tangent bundle $TM \to M$ and for a smooth manifold N, it assigns the the smooth map $F : N \to M$ the differential or the pushforward $F_* : TM \to TN$. The case of 1-forms is just dual to this case. The cotangent functor T^* is a contravariant functor that assigns to each smooth map $F : N \to M$ the codifferential or the pullback

$(F_*)^\vee : T^*M \to T^*N$. Since, the concept of codifferential is new to us, let us look at it more explicitly:

The codifferential, or the dual of a differential,

$$(F_{*,p})^\vee : T^*_{F(p)} \to T^*_p N$$

reverses the arrows and pulls back a covector at $F(p)$ from M to N. For simplicity, we denote the codifferential by F^*.

One of the added advantages that forms give us over vector fields is that vector fields cannot be push forwarded under C^∞ maps, but every covector field can be pulled back by a C^∞ map. Under this codifferential map, if ω is a 1-form defined on M, then, $F^*(\omega)$ is a 1-form defined on N pointwise by

$$(F^*\omega)_p = F^*(\omega_{F(p)}), \ \mathrm{p} \in N.$$

That is, $\forall X_p \in T_pN$, we have:

$$(F^*\omega)_p X_p = \omega_{F(p)}(F_*(X_p)).$$

By definition, $C^\infty(M)$, that is, the set of all smooth functions over M are said to be the 0-forms over the manifold M. Hence, under this map F^*, any $g \in C^\infty(M)$ can also be pulled back: $F^*g = g \circ F \in C^\infty(N)$.

Now, a natural question to ask is whether a C^∞ 1-form under a C^∞ map C^∞? We answer this question by establishing first the following commutation properties of the pullback:

Proposition 1.11.6. ***(Commutation of the pullback with the differential).*** *Let $F : N \to M$ be a C^∞ map of manifolds. For a $h \in C^\infty(M)$, $F^*(dh) = d(F^*h)$.*

Proof. □

Proposition 1.11.7. ***(Pullback of a sum and product).*** *Let $F : N \to M$ be a C^∞ map. Then for $\omega, \tau \in \Omega^1(M)$, and $g \in C^\infty(M)$ we have:*

1. $F^*(\omega + \tau) = F^*(\omega) + F^*(\tau)$,
2. $F^*(g\omega) = (F^*g)(F^*\omega)$.

Proof. □

Proposition 1.11.8. *The pullback $F^*\omega$ of a C^∞ 1-form ω under a C^∞ map $F : N \to M$ is C^∞.*

Proof. □

1.12 Differential k-forms

We now generalize the notion of differential 1-forms to differential k-forms. Consider the vector bundle $\bigwedge^k(T^*M)$, which is called the k-the exterior power of the cotangent bundle, or the exterior bundle, which is the space of all alternating k-tensors on the tangent bundle TM. A differential k-form ω is a k-covector field which assigns to each point $p \in M$ a k-covector $\omega_p \in \bigwedge^k(T_p^*)$, and this form ω is smooth if it is smooth as a section of the exterior bundle $\bigwedge^k(T^*)$.

A *top form* on a manifold is a differential form whose degree is same as the dimension of the manifold.

Lemma 1.12.1. ***Multilinearity of a form over functions***. *For any vector fields $X_1, \dots, X_k$, and any function h on M,*

$$\omega(X_1, \dots, hX_i, \dots, X_k) = h\omega(X_1, \dots, X_i, \dots, X_k).$$

where the function $\omega(X_1, \dots, X_i, \dots, X_k)$ is defined pointwise for each point $p \in M$.

1.12.1 Local expression of a differential k-form

Let $(U, \phi, x_1, \dots, x_n)$ be a coordinate chart and $p \in M$ be an arbitrary point. Let $\frac{\partial}{\partial x_1}\Big|_p, \dots, \frac{\partial}{\partial x_n}\Big|_p$ be a basis of T_pU and let $dx_1|_p, \dots, dx_n|_p$ be its dual basis, that is, the basis for the cotangent space T_p^*U. Then a basis for $\bigwedge^k(T_p^*U)$ is given by $\{(dx_{i_1})|_p \wedge \dots \wedge (dx_{i_k})|_p\}_{1 \leq i_1 < i_2 < \dots < i_k \leq n}$.

Certain nice exercises:

1) We try to show the dimension of a connected topological manifold is well defined using the following steps:

Step 1: We first define a function $f : M \to \mathbb{N}$ in the following prescription:

Let $p \in M$. If there is a coordinate chart (U, ϕ) around p, such that $\phi(U) \subset \mathbb{R}^n$, then define $f(p) = n$. We show that f is open using the fact that if U and V are homeomorphic spaces such that U and V are open subsets of $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively, then $m = n$.

Let us choose a maximal atlas $A = \{(U_\alpha, \phi_\alpha)\}_{\alpha \in \mathcal{A}}$ for the topological manifold M, and we know that the same forms a basis for the topology on the manifold M. So, to prove that f is an open map, it is enough to show that $f(U_\alpha)$ is open in $\mathbb{N}$, where (U_α, ϕ_α) is a coordinate chart about a point, say p. Since, $\mathbb{N}$ has discrete topology, all the open subsets of $\mathbb{N}$ are the singletons, *i.e.*$\{n\}$, where $n \in \mathbb{N}$. So, for $f(U_\alpha)$ to be open, it should be one of these singleton subsets of $\mathbb{N}$. If that's not the case, then $\exists\ m \in f(U_\alpha)$ and a chart $\psi : U_\alpha \to \mathbb{R}^m$ such that $\psi \circ \phi^{-1} : \phi(U_\alpha) \to \psi(U_\alpha)$ is a homeomorphism, where $\phi(U_\alpha)$ and $\psi(U_\alpha)$ are two open subsets of $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively, from which we can conclude that $n = m$. Hence, $f(U_\alpha) = \{n\}$, and thus f is open.

Step 2: We now show that f is a continuous map which is locally constant.

It is enough to check the continuity of f on a basis element of $\mathbb{N}$. So, let $n \in \mathbb{N}$ be arbitrary. It suffices to show that $f^{-1}(\{n\})$ is open. But, $f^{-1}(\{n\}) = \{p \in M : \exists$ a coordinate chart (U, ϕ) containing p such that $\phi(U_\alpha) \subset \mathbb{R}^n\}$. Let, $\mathcal{U} := \bigcup\limits_{(U_\alpha, \phi_\alpha) \in A} \{U_\alpha : p \in U_\alpha \text{ and } \phi(U_\alpha) \subset \mathbb{R}^n\}$. We claim that $f^{-1}(\{n\}) = \mathcal{U}$. Since, $\mathcal{U}$ is the union of coordinate charts, $\mathcal{U}$ is open, and thus proving our claim would imply that f is continuous. It is clear that $f^{-1}(\{n\}) \subset \mathcal{U}$. Now let, $q \in \mathcal{U}$. Then, $\exists$ a coordinate chart $(U_\alpha, \phi_\alpha) \in \mathcal{U}$ which contains q. And, clearly by definition of $\mathcal{U}$, $\phi(U_\alpha) \subset \mathbb{R}^n$ which implies $f(q) = n$ and hence, $q \in f^{-1}(\{n\})$. Thus, $\mathcal{U} \subset f^{-1}(\{n\})$.

Also, it is easy to see that f is locally constant. Since, by hypothesis, for every point, say p contained in a coordinate chart, say (U_α, ϕ_α), (which as mentioned earlier, is an element of a basis of the manifold M), $\phi_\alpha(p) = n$ for some $n \in \mathbb{N}$.

Step 3: We now use the connectedness of our given manifold M to finish the proof.

Since, M is connected, it cannot be be written as a disjoint union of two non-empty open subsets of M, or we can say that M cannot be written as a disjoint union of coordinate charts of A. Since, f is locally constant on M, it takes a constant value in $\mathbb{N}$ in each of these coordinate chart. Now, since none of these charts are disjoint, f agrees in the intersection of all these coordinate charts, and since these coordinate charts cover the entire manifold M, it follows that f takes a constant value in $\mathbb{N}$ on the entire manifold M.

Hence, for each point p in M, $\exists$ a coordinate chart (U_α, ϕ_α) such that $\phi_\alpha(U_\alpha) \subset \mathbb{R}^n$ for some $n \in \mathbb{N}$. Hence, the dimension of a topological manifold is a well defined notion.

2) We show that two smooth atlas of a topological manifold determine the same maximal atlas if and only if their union is again a smooth atlas.

Let A and B be two smooth atlas of a topological manifold M and let, $\bar{A}$ and $\bar{B}$ denote the maximal atlas of A and B respectively. We prove that $\bar{A} = \bar{B}$ if and only if $A \cup B$ is a smooth atlas.

The implication $(\Rightarrow)$ is trivial, since, $\bar{A} = \bar{B}$ implies $\bar{A}(= \bar{B})$ is a smooth maximal atlas containing both A and B, that is every coordinate chart of A is compatible with every coordinate chart of B, and hence $A \cup B$ is a smooth atlas.

To prove ($\Leftarrow$) let $A \cup B$ be a smooth atlas. We show $\bar{A} = \bar{B}$.
Now, $\bar{A} :=$ {collection of all coordinate charts of M such that they are compatible with coordinate charts of A}. To prove the required, we choose a coordinate chart (U_A, ϕ_A) (note: this chart may not be contained in A) from $\bar{A}$ and a coordinate chart (V_B, ψ_B) from B and show that the transition functions $\psi_B \circ \phi_A^{-1}$ and $\phi_A \circ \psi_B^{-1}$ are smooth. Since $A \cup B$ is smooth, every coordinate chart of A is compatible with every coordinate chart of B. Now, if we select any coordinate chart (W_A, f_A) from the atlas A, then $\psi_B \circ f_A^{-1}$ is smooth. Hence, the transition function $\psi_B \circ \phi_A^{-1}$ being the composition of smooth functions $\psi_B \circ f_A^{-1}$ and $f_A \circ \phi_A^{-1}$, is smooth. Similarly, $\phi_A \circ \psi_B^{-1}$ is also smooth. Hence, we can conclude that every coordinate chart of $\bar{A}$ is contained in $\bar{B}$. In a similar way, we can show that every coordinate chart of $\bar{B}$ is contained in $\bar{A}$ by considering coordinate charts from $\bar{B}$ and A and showing that their transitions functions are smooth. Hence, $\bar{A} = \bar{B}$.

3) We try to give a smooth structure on n-dimensional real projective space $\mathbb{R}P^n$.

First, let us show that $\mathbb{R}P^n$ is second countable and Hausdorff.

To show that $\mathbb{R}P^n$ is second countable, we recall a very basic fact that, for any second countable topological space X equipped with an equivalence relation $\sim$, if the natural map $\pi : X \to X/\sim$ is open, then the quotient space $X/\sim$ is second countable. Since, $\mathbb{R}^{n+1} \setminus \{0\}$ is second countable, it suffices to show that the natural map $\pi : \mathbb{R}^{n+1} \setminus \{0\} \to \mathbb{R}P^n$ is open. Let U be an open subset of $\mathbb{R}^{n+1} \setminus \{0\}$. Now, for any $t \in \mathbb{R} \setminus \{0\}$, the set

$$tU = \{tx : x \in U\}$$

is an open subset of $\mathbb{R}^{n+1} \setminus \{0\}$. Therefore,

$$\pi^{-1}(\pi(U)) = \bigcup_{t \in \mathbb{R} \setminus \{0\}} tU$$

is open. Hence, $\pi(U)$ is open in $\mathbb{R}P^n$ by definition of quotient topology.

To show that $\mathbb{R}P^n$ is Hausdorff, we consider the quotient map $q : S^n \to \mathbb{R}P^n$ (it is convinient to work with the domain being S^n instead of $\mathbb{R}^{n+1} \setminus \{0\}$ because S^n is embedded in $\mathbb{R}^{n+1} \setminus \{0\}$ and to obtain the quotient space $\mathbb{R}P^n$, all that we are doing is to identify the anti-podal points of S^n). Let $u, v \in \mathbb{R}P^n$ with $u \neq v$. Then, there are $x, y \in S^n$ such that $q^{-1}([u]) = \{x, -x\}$ and $q^{-1}([v]) = \{y, -y\}$. Let, $\epsilon = \frac{1}{2}\min\{\|x-y\|, \|x+y\|\}$ and set $U = B(x,\epsilon) \cap S^n$ and $V = B(y,\epsilon) \cap S^n$ where the open balls are taken in $\mathbb{R}^{n+1}$. Then, $U, V, -U, -V$ are pairwise disjoint open neighbourhoods of $x, y, -x, -y$ respectively in S^n. Moreover, $q^{-1}(q(U)) = U \sqcup -U$ and $q^{-1}(q(V)) = V \sqcup -V$. Then, it is easy to see that $q(U)$ and $q(V)$ are disjoint open neighbourhoods of u and v respectively in $\mathbb{R}P^n$.

Now, we proceed on to give a smooth structure to $\mathbb{R}P^n$. We cover $\mathbb{R}P^n$ with homogeneous coordinate charts (U_i, ϕ_i) where $U_i = \{[a^0, \dots, a_i, \dots, a^n] \in \mathbb{R}P^n : a^i \neq 0\}$ and the coordinate functions are defined as:

$$\begin{aligned} \phi_i &: U_i \to \mathbb{R}^n \\ [a^0, \dots, a_i, \dots, a^n] &\mapsto \Big(\frac{a^0}{a^i}, \frac{a^1}{a^i}, \dots, \frac{a^n}{a^i}\Big) \forall i = 1, \dots, n. \end{aligned}$$

This map admits a continuous inverse defined as:

$$\begin{aligned} \phi_i^{-1} &: \phi_i(U_i) \to \mathbb{R}P^n \\ (b^1, \dots, b^n) &\mapsto [b^1, b^2, \dots, b^{i-1}, 1, b^{i+1}, \dots, b^n] \end{aligned}$$

Clearly, ϕ_i is a homeomorphism for each $i = 1, 2, \ldots, n$. Hence, $\mathbb{R}P^n$ is locally Euclidean with (U_i, ϕ_i) as charts.

Now, to prove that the collection of these charts give a smooth structure to $\mathbb{R}P^n$, that is, they define a smooth atlas for $\mathbb{R}P^n$, we are just left to show that the transition functions defined on the intersection of any two charts from this collection is also smooth.

To prove this, WLOG, we consider the charts (U_0, ϕ_0) and (U_1, ϕ_1), where on the intersection of these two charts, we have $a_0, a_1 \neq 0$. Now, since there are two coordinate systems, we refer to the coordinate functions on U_0 as $x_1, \ldots, x_n$ and the coordinate functions on U_1 as $y_1, \ldots, y_n$.

$$\text{On } U_0, x^i = \frac{a^i}{a^0}, i = 1, 2, \ldots, n.$$
$$\text{On } U_1, y^i = \frac{a^i}{a^1}, i = 0, 2, \ldots, n.$$

Then solving these two coordinate systems, we obtain on $U_0 \cap U_1$,

$$y^1 = \tfrac{1}{x^1}; y^2 = \tfrac{x^2}{x^1}; \ldots, y^n = \tfrac{x^n}{x^1}.$$

And clearly, the transition function,

$$\begin{aligned} \phi_1 \circ \phi_0^{-1} : \phi_0(U_0 \cap U_1) &\to \phi_1(U_0 \cap U_1) \\ (x^1, \ldots, x^n) &\mapsto \left(\frac{1}{x^1}, \frac{x^2}{x^1}, \ldots, \frac{x^n}{x^1}\right) \end{aligned}$$

is smooth since $x_1 \neq 0$ on $\phi_0(U_0 \cap U_1)$. Similarly, we can easily compute $\phi_0 \circ \phi_1^{-1}$ to find that it's smooth. On any other $U_i \cap U_j$, an analogous formula holds. Therefore, the collection $\{(U_i, \phi_i)\}_{i=0,\ldots,n}$ defines a smooth structure on $\mathbb{R}P^n$, and we also call it the standard or smooth atlas on $\mathbb{R}P^n$. Henceforth, $\mathbb{R}P^n$ is a smooth manifold.

4) Let $M_1, \ldots, M_k$ be smooth manifolds of dimension $n_1, \ldots, n_k$ respectively. We show that $M_1 \times \cdots \times M_k$ is a smooth manifold of dimension $(n_1 + \cdots + n_k)$.

Part II

Foundational Algebraic Topology

Part III

Surgery on Manifolds

Chapter 2

Classification Of Manifolds

Manifolds are good. But you know what's better than manifolds? Their classification. This is the content of this part, where we will classify manifolds of dimension 0,1,2,3, see the obstruction to a *nice* classification of 4 manifolds and will begin the surgery method for classifying higher manifold (namely, manifolds of dimension ≥ 5). In this part of the note, we would like to work only with differentiable, compact and closed manifolds, *unless otherwise specified.*

There is a unique 0-dimensional manifold, namely the point. Hence, all other disconnected 0-dimensional manifolds are just discrete sets, which can be only studied by methods of combinatorics, as they have no geometry.

One dimensional manifolds are also similarly easy to classify. They are homeomorphic to the circle, if compact, otherwise, they are homeomorphic to the real line. Every one dimensional manifold is just a curve embedded in some higher dimensional Euclidean space, and by methods of topology, such as one point compactification of the real line, the aforesaid is very easy to observe.

Manifolds of dimension 2 and 3 can be classified using geometric tools.

The study of 2-dimensional manifolds (surfaces) is deeply connected with complex analysis and algebraic geometry because every orientable (since, every connected 2-dimensional manifold admits a constant curvature metric, that is, it has curvature either positive, negative or zero) surface can be considered a Riemann surface [1] or a complex algebraic curve. Via these tools, one can observe that every surface is homeomorphic to either a sphere, torus, connected sum of tori, projective plane, or connected sum of projective planes. Using tools from algebraic topology it is not that much of a hardship to show that these surfaces are not homeomorphic to one another. Hence, these 2-manifolds can be classified into exactly one of these equivalence classes.

[1] Riemann surfaces are connected one dimensional complex manifolds.